Engineering Mathematics – V : Probability & Random Processes

Unit III : Random Processes

Classification, Stationarity, Correlation Functions and Power Spectral Density

Dr. S. Balaji

Contents

1 Random Processes: Basics **3**

1.1 What is a random process? . . . 3
1.2 Two motivating examples . . . 4
1.3 Classification of random processes . . . 5
1.4 Stationary random processes . . . 8
1.5 Correlation and the autocorrelation function . . . 9
1.6 Wide-sense stationary (WSS) process . . . 11
1.7 Summary . . . 13

2 Stationary Processes and Correlation Functions **14**

2.1 Strict-Sense Stationary (SSS) Process . . . 14
2.2 Wide-Sense Stationary (WSS) Process . . . 14
2.3 Definition: Uniform Distribution . . . 14
2.4 Problem 1 – Non-Stationary Random Process . . . 15
2.5 Problem 2 – Stationarity of $X(t) = \cos(\omega_0 t + \theta)$ . . . 16
2.6 Problem 3 – Stationarity of $X(t) = A\cos(\omega_0 t + \theta)$ . . . 17
2.7 Example – Stationarity of $X(t) = A\sin(\omega t + \phi)$ . . . 18
2.8 Problem – $X(t) = A\cos(\omega_0 t + \theta)$ is a WSS Process . . . 19
2.9 Problem – $Y(t) = X(t)\cos(\omega t + \phi)$ is a WSS Process . . . 20
2.10 Exercise . . . 23
2.11 Problem – $X(t) = A\cos\omega t + B\sin\omega t$ is a WSS Process . . . 23
2.12 Problem – $X(t) = Y\cos\omega t + Z\sin\omega t$ is a WSS Process . . . 24
2.13 Exercise . . . 25
2.14 Definitions . . . 25
2.15 Autocorrelation Function (ACF) – Definition . . . 25
2.16 Properties of the ACF – Property 1 . . . 25
2.17 Properties of the ACF – Property 2 . . . 26
2.18 Properties of the ACF – Property 3 . . . 26
2.19 Problem 1 – Suitable ACF . . . 27

2.20 Problem 2 – Mean and Variance from ACF . . . 27
2.21 Problem 3 – Mean, Variance and Covariance . . . 27
2.22 Problem 4 – Second Moment of $X(8) - X(5)$ . . . 28
2.23 Cross-Correlation Function (CCF) – Property 1 . . . 28
2.24 CCF – Property 2 . . . 29
2.25 CCF – Property 3 . . . 29
2.26 CCF – Property 4 . . . 30
2.27 Problem – CCF of $W(t)$ and $Z(t)$ . . . 30
2.28 Problem – CCF of Two Sinusoids . . . 31

3 Power Spectral Density 33
3.1 Introduction . . . 33
3.2 Definition . . . 33
3.3 Properties of the PSD . . . 34
3.4 Wiener–Khinchine Theorem . . . 35
3.5 Wiener–Khinchine Theorem – Proof . . . 35
3.6 Problems . . . 38
3.7 PSD of the Random Telegraph Signal . . . 47
3.8 Summary . . . 49

4 Assignment – 4 50
4.1 Questions . . . 50
4.2 Hints . . . 52

1 Random Processes: Basics

Definition, Classification, Stationarity and the Autocorrelation Function

1.1 What is a random process?

Random Process (RP)

A **random process** is a *collection of random variables* (or a *collection of random signals*).

- A **random variable** involves **only** the sample space S: $X = X(s)$, $s \in S$.
- A **random process** involves **both** the sample space S and a parameter (time) set T: $X = X(t, s)$, $t \in T$, $s \in S$.

Usually the argument s is suppressed and we simply write $\{X(t)\}$.

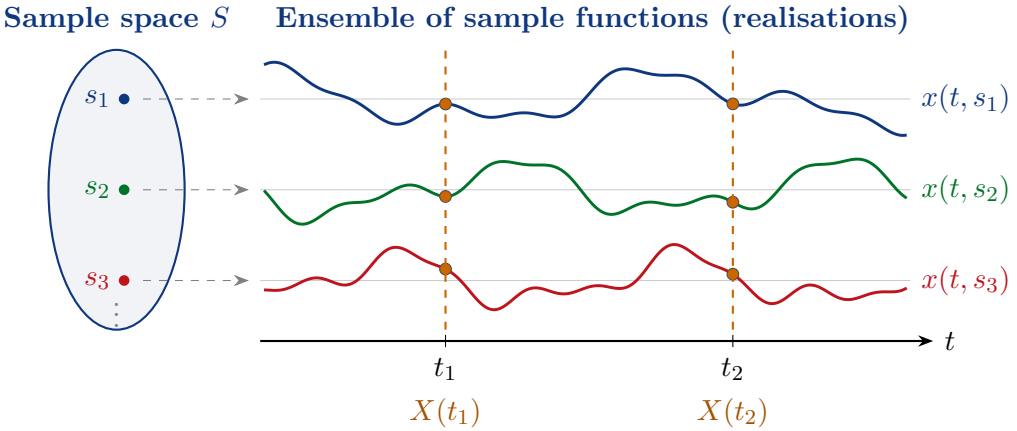
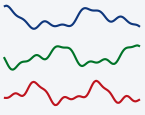


Figure 1: A random process as a collection of signals. Fix t and look across the ensemble: each vertical slice is a random variable.

Four ways to read $X(t, s)$

t varies, s varies



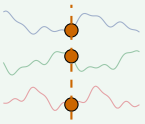
A family of functions: **the random process** itself.

t varies, s fixed



A single time function: a **sample function** (realisation).

t fixed, s varies



A **random variable** $X(t_1)$.

t fixed, s fixed



Just a **real number** $x(t_1, s_1)$.

1.2 Two motivating examples

Example 1: arrivals at a railway junction

Example 1

$X(t)$ = number of people arriving at a railway junction in the time interval $(0, t)$ hours.

- $t = 2$: $X(2)$ is one random variable,
- $t = 3.5$: $X(3.5)$ is another random variable,
- $t = 10$: $X(10)$ is yet another random variable.

This number of arrivals may be represented through $(1 + at)^n$; e.g.

$$P\{X(t) = n\} = \frac{(at)^{n-1}}{(1 + at)^{n+1}}, \qquad n = 1, 2, 3, \ldots$$

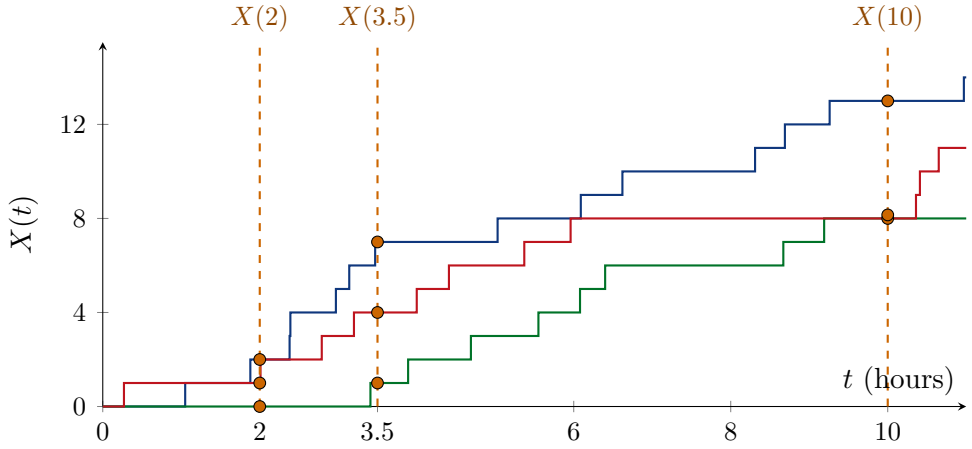


Figure 2: Three sample functions of the arrival process. At each fixed time instant the ensemble of values forms a random variable.

Example 2: signals of a radio station

Example 2

Signals of a radio station received by different receivers, lying at different distances, may be represented as a random process

$$X(t) = A\cos(\omega t + \phi).$$

Here the amplitude A and the phase ϕ vary with the distance of the receiver from the radio station; hence A **and** ϕ **are random variables** (similarly t may also be treated as a random variable).

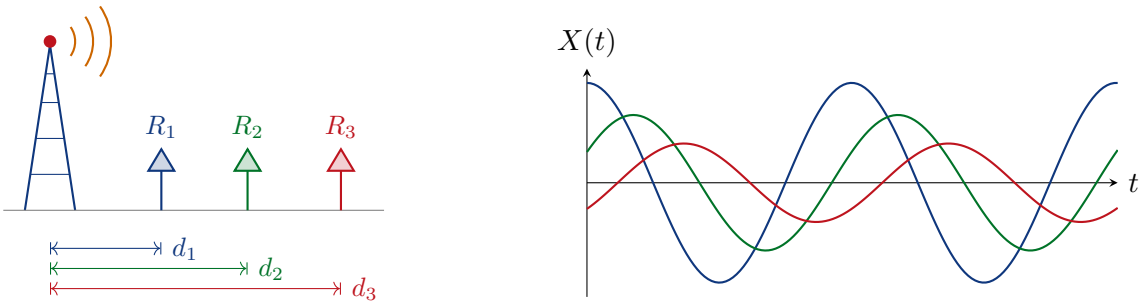


Figure 3: Receivers R_1, R_2, R_3 at distances $d_1 < d_2 < d_3$ (left) and the signals they receive (right): a farther receiver gives a different amplitude A and phase ϕ.

1.3 Classification of random processes

Based on the nature of the parameter set T and the sample (state) space S, we have the following classification of random processes.

	T discrete	T continuous
S discrete	① Discrete random sequence	③ Continuous random sequence
S continuous	② Discrete random process	④ Continuous random process

Figure 4: The four types of random processes.

Memory aid

T decides **discrete / continuous**; S decides **sequence / process**.

Type 1: both T and S discrete

Discrete random sequence

If both T and S are discrete, the RP $X(t, s)$ is called a **discrete random sequence**.

Example

Number of child births in the time interval of every one hour, in a metropolitan city.

$$S = \{0, 1, 2, 3, 4, \dots\}, \qquad T = \{1, 2, 3, \dots, 24\}.$$

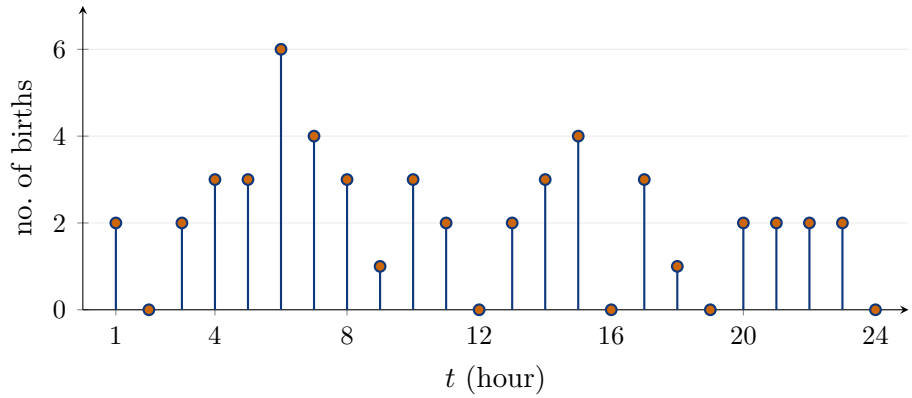


Figure 5: Type 1: isolated time points, integer values.

Type 2: T discrete and S continuous

Discrete random process

If T is discrete and S is continuous, the RP $X(t, s)$ is called a **discrete random process**.

Example

The temperature recorded at a place in every 4 hours of duration.

$$S = \{15.8°\text{C} - 40.6°\text{C}\}, \qquad T = \{4, 8, 12, 16, 20, 24\}.$$

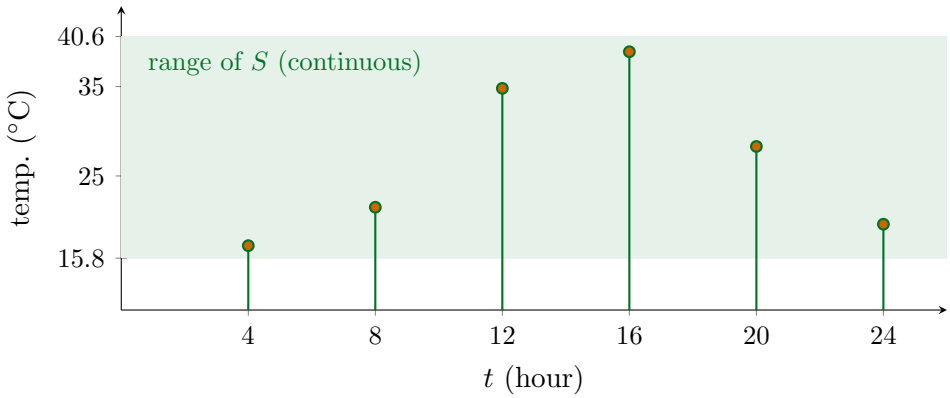


Figure 6: Type 2: isolated time points, any real value in a range.

Type 3: T continuous and S discrete

Continuous random sequence

If T is continuous and S is discrete, the RP $X(t, s)$ is called a **continuous random sequence**.

Example

Number of customers arriving at a counter in the time interval of $(0, t)$ minutes; $S = \{0, 1, 2, \dots\}$, $T = [0, \infty)$.

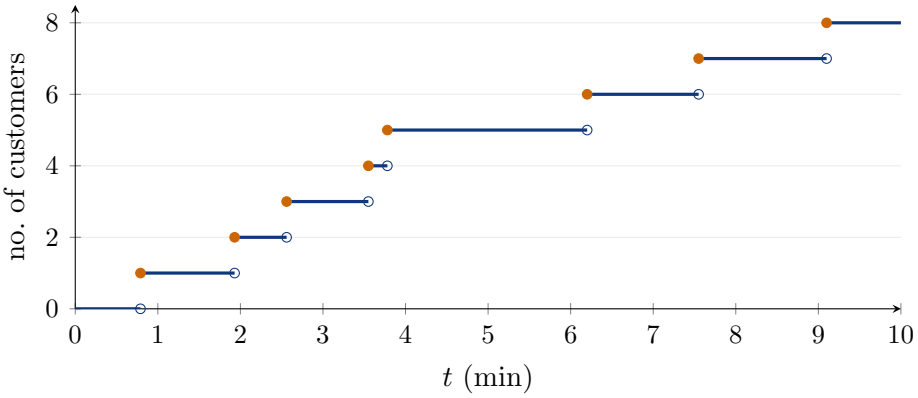


Figure 7: Type 3: defined for every t, jumps by integer steps.

Type 4: both T and S continuous

Continuous random process

If both T and S are continuous, the RP $X(t, s)$ is called a **continuous random process**.

Example

Maximum temperature recorded at a place in the time interval of $(0, t)$ hours; S is an interval of real numbers and $T = [0, \infty)$.

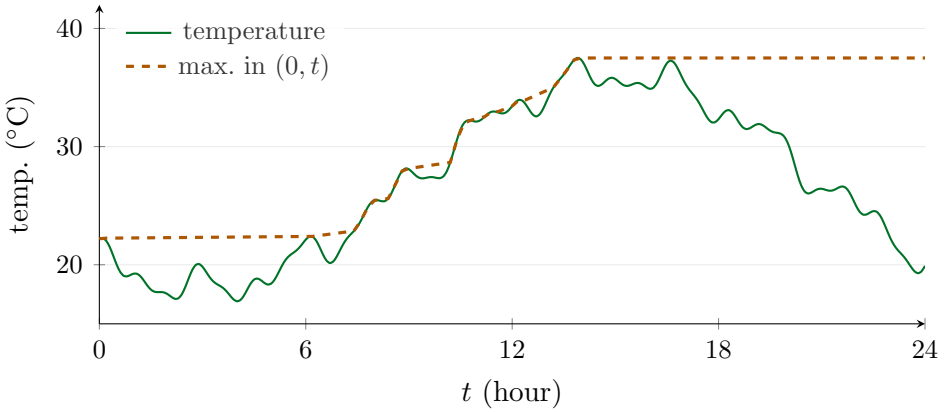


Figure 8: Type 4: defined for every t and taking any real value; the dashed curve is the maximum temperature recorded in $(0, t)$.

1.4 Stationary random processes

Stationary RP

A random process $X(t, s)$ whose distributions (values) do not vary with respect to the time parameter t is called a **stationary** random process.

Strictly stationary RP

A random process $\{X(t)\}$ is said to be **strictly stationary** if it does not vary with respect to the time t.

How do we check?

Compute the **statistical averages** of $X(t)$ – such as the **mean** and the **variance** – and see whether they depend on t.

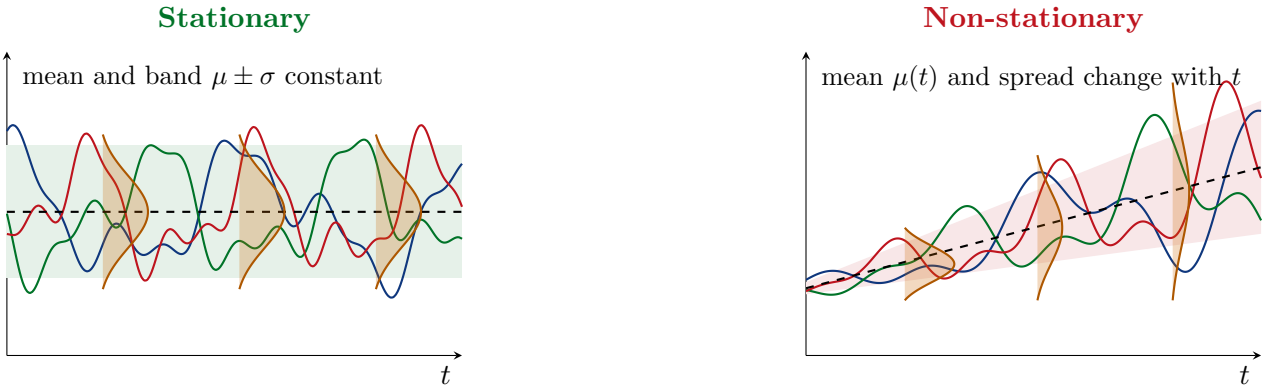


Figure 9: Amber curves: the distribution of $X(t)$ at three time instants – identical (left), shifting and spreading (right).

Strict-sense stationary (SSS) process

SSS / strongly stationary process

If both the mean and the variance values of a random process $\{X(t)\}$ are constant, then it is called a **strict-sense stationary** (SSS) or **strongly stationary** process. That is, $\{X(t)\}$ is an SSS process if

(i) mean of $X(t) = \mathbb{E}[X(t)] =$ a constant,

(ii) $\mathrm{Var}[X(t)] =$ a constant,

i.e. neither is a function of time t.

Remark (formal view)

Formally, strict-sense stationarity means that *every* joint distribution of $X(t_1), X(t_2), \ldots, X(t_n)$ is unchanged when all the time instants are shifted by

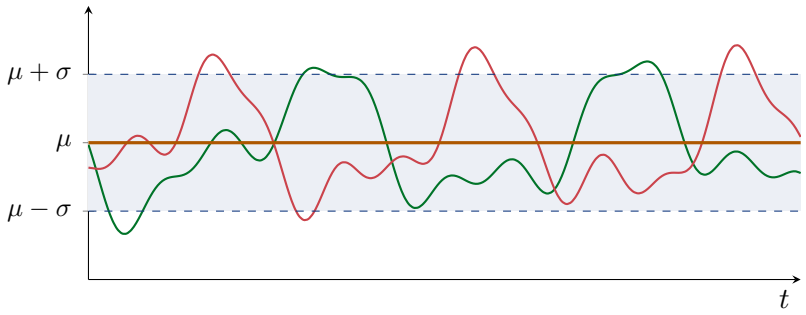


Figure 10: For an SSS process the mean line and the $\pm\sigma$ band are flat.

the same amount c:

$$F(x_1,\ldots,x_n;\, t_1+c,\ldots,t_n+c) = F(x_1,\ldots,x_n;\, t_1,\ldots,t_n).$$

Constant mean and constant variance are the working checks used while solving problems.

1.5 Correlation and the autocorrelation function

Correlation between two random variables

If X and Y are two random variables, then the **correlation** between X and Y is given by $\mathbb{E}(XY)$.

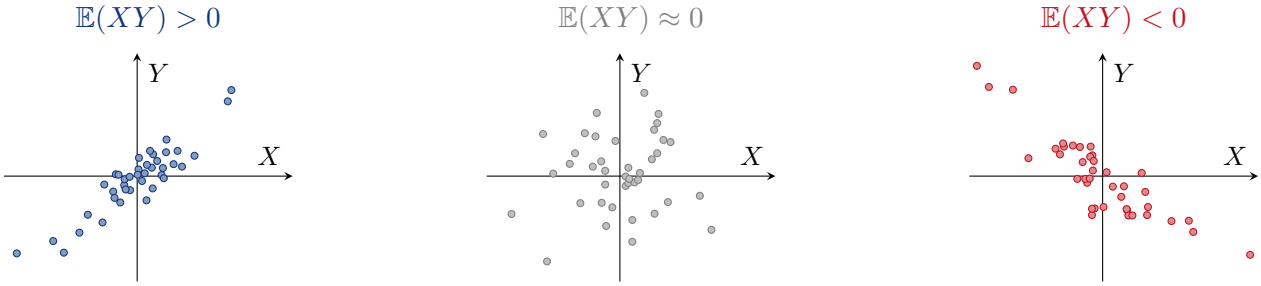


Figure 11: For zero-mean X and Y, $\mathbb{E}(XY)$ measures how strongly the two variables move together.

Autocorrelation function (ACF)

Autocorrelation function

If $\{X(t)\}$ is a random process, then its **autocorrelation function** (ACF) is defined by

$$R_{XX}(t_1,t_2) = \mathbb{E}\big[X(t_1)\,X(t_2)\big].$$

For a stationary random process $\{X(t)\}$, the ACF will be a function of the time difference $t_2 - t_1$ (or $t_1 - t_2$).

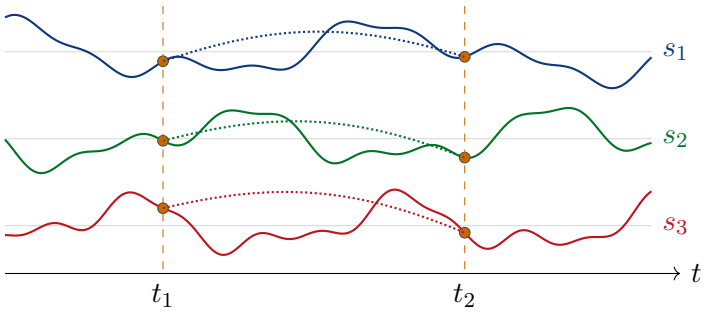


Figure 12: The ACF as an ensemble average of the products $X(t_1)X(t_2)$.

Multiply the pair of values on each sample function and **average over the ensemble**:

$$R_{XX}(t_1, t_2) \approx \frac{1}{N} \sum_{k=1}^{N} x(t_1, s_k)\, x(t_2, s_k).$$

Notation and the stationary case

The ACF is generally denoted by $R_{XX}(t_1, t_2)$ or $R_X(t_1, t_2)$ or $R(t_1, t_2)$, so that

$$R_X(t_1, t_2) = \mathbb{E}\big[X(t_1)\, X(t_2)\big].$$

For a stationary RP

$$R_X(t_1, t_2) = \phi(t_1 - t_2),$$

a **function of the time difference** $\tau = t_1 - t_2$ (or $t_2 - t_1$) only.

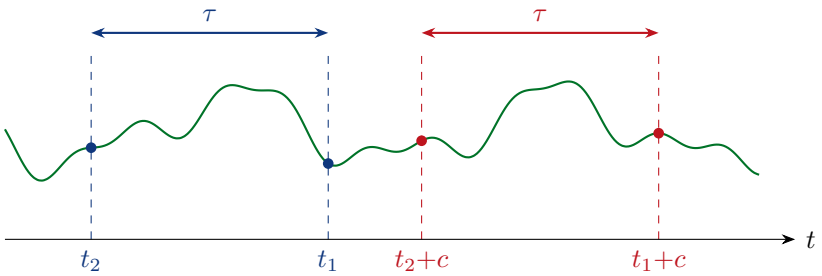


Figure 13: The same gap τ anywhere on the time axis gives the same value of R for a stationary random process.

Significance of the ACF

Note

The main significance of the ACF is that the **expected frequencies of the signal** in the next two time instants can be identified.



Figure 14: Two typical autocorrelation functions: the peak occurs at $\tau = 0$, where $R(0) = \mathbb{E}[X^2(t)]$ is the mean power, and R is symmetric about $\tau = 0$.

A slowly varying (low-frequency) signal stays correlated over a long time gap, so its ACF is broad; a rapidly varying (high-frequency) signal loses correlation quickly, so its ACF is narrow.

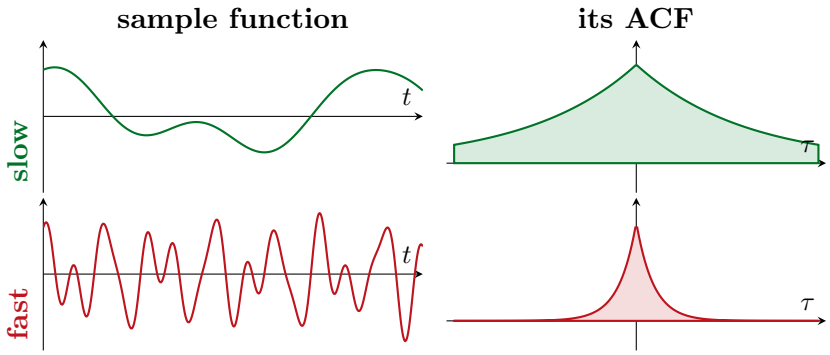


Figure 15: Slowly varying (low-frequency) signal ⇒ broad ACF; rapidly varying (high-frequency) signal ⇒ narrow ACF.

1.6 Wide-sense stationary (WSS) process

WSS process

A random process $\{X(t)\}$ is said to be a **wide-sense stationary** (WSS) process if

(i) $\mathbb{E}[X(t)]$ = a constant,

(ii) the ACF $R_X(t_1, t_2)$ is a function of the time difference,

i.e. $R_X(t_1, t_2) = R_X(\tau)$ with $\tau = t_1 - t_2$.

SSS and WSS side by side

SSS process

- $\mathbb{E}[X(t)]$ = constant
- $\mathrm{Var}[X(t)]$ = constant

(formally: all joint distributions are shift-invariant)

WSS process

- $\mathbb{E}[X(t)]$ = constant
- $R_X(t_1, t_2) = R_X(t_1 - t_2)$

(only first- and second-order moments)

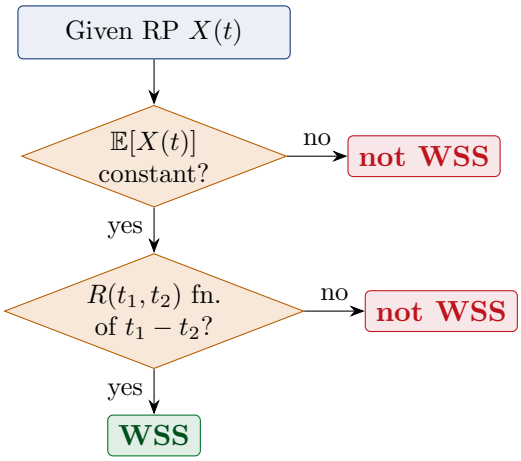


Figure 16: Testing a random process for wide-sense stationarity.

For a WSS process the variance is automatically constant:

$$\mathrm{Var}[X(t)] = \mathbb{E}[X^2(t)] - \mu^2 = R_X(0) - \mu^2.$$

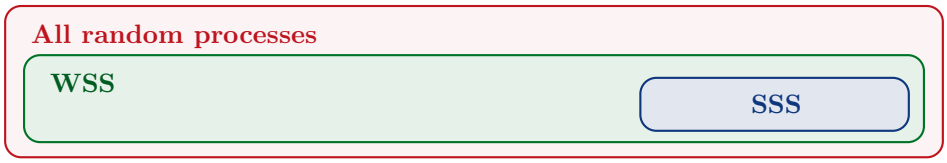


Figure 17: With finite second moments, every SSS process is also WSS; the converse need not hold.

Illustration: the random-phase cosine

Back to Example 2

Let $X(t) = A\cos(\omega t + \phi)$, where A and ω are constants and ϕ is uniformly distributed in $(0, 2\pi)$.

Mean.

$$\mathbb{E}[X(t)] = \frac{A}{2\pi}\int_0^{2\pi} \cos(\omega t + \phi)\, d\phi = 0 \quad \text{(a constant)}.$$

Autocorrelation function.

$$\begin{aligned} R(t_1, t_2) &= A^2\, \mathbb{E}\big[\cos(\omega t_1 + \phi)\cos(\omega t_2 + \phi)\big] \\ &= \frac{A^2}{2}\, \mathbb{E}\big[\cos\omega(t_1 - t_2) + \cos\big(\omega(t_1 + t_2) + 2\phi\big)\big] = \frac{A^2}{2}\cos\omega(t_1 - t_2), \end{aligned}$$

which is a function of the time difference $\tau = t_1 - t_2$ alone. Hence $\{X(t)\}$ is a **WSS process**.

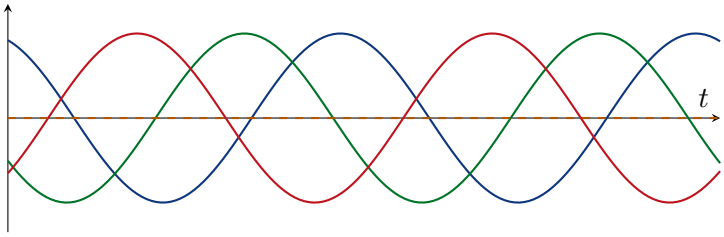


Figure 18: Sample functions of $X(t) = A\cos(\omega t + \phi)$ for three values of the random phase ϕ; the ensemble mean (dashed) is zero at every t.

1.7 Summary

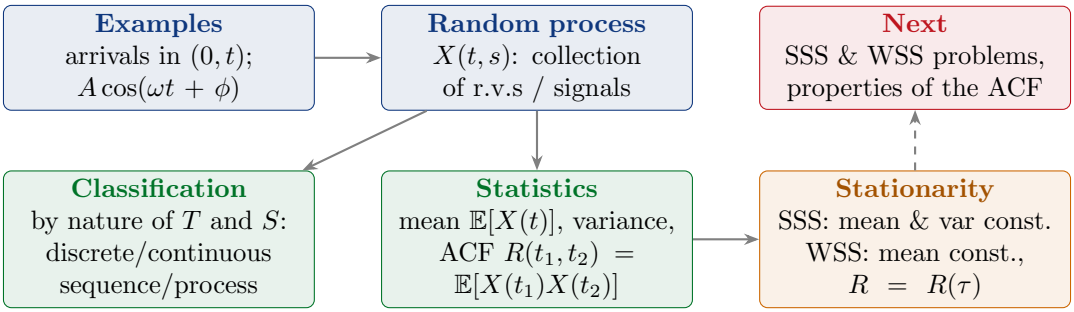


Figure 19: Concept map of Unit III – random process basics.

Quick revision

1. RP = collection of random variables; $X(t, s)$ involves both S and T.
2. T decides discrete / continuous; S decides sequence / process.
3. Stationary: statistical averages do not change with t.
4. SSS: mean and variance are constants.
5. ACF: $R_X(t_1, t_2) = \mathbb{E}[X(t_1)X(t_2)]$; for a stationary RP it depends only on $t_1 - t_2$.
6. WSS: mean is a constant *and* the ACF is a function of the time difference.

2 Stationary Processes and Correlation Functions

SSS and WSS Processes, Autocorrelation and Cross-Correlation

2.1 Strict-Sense Stationary (SSS) Process

Strict-Sense Stationary (SSS) Process

A random process $\{X(t)\}$ is **SSS** if its statistical properties remain unchanged by any shift c of the time origin, i.e. for all n and c,

$$f(x_1, \ldots, x_n; t_1, \ldots, t_n) = f(x_1, \ldots, x_n; t_1 + c, \ldots, t_n + c).$$

Consequently,

$$(i) \quad E[X(t)] = \text{constant}, \qquad (ii) \quad \text{Var}[X(t)] = \text{constant}.$$

In the problems below, (i) and (ii) are used as the working test for stationarity.

2.2 Wide-Sense Stationary (WSS) Process

Wide-Sense Stationary (WSS) Process

A random process is **WSS** if only its first and second-order moments are time-invariant:

$$(i) \quad E[X(t)] = \text{constant},$$
$$(ii) \quad R_X(t_1, t_2) = R_X(t_1 - t_2) = R_X(\tau), \qquad \tau = t_1 - t_2,$$

i.e. R_X depends only on the time difference τ.

A random process $\{X(t)\}$ which is not stationary is called an **evolutionary process**.

2.3 Definition: Uniform Distribution

Uniform Distribution

A random variable X is said to follow a **uniform distribution** in (a, b) if its pdf is

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

Then X is uniformly distributed on (a, b).

2.4 Problem 1 – Non-Stationary Random Process

Question

Customers arrive at a counter during time t minutes. Let $\{X(t)\}$ be a random process with probability distribution

$$P(X(t)=n)=\begin{cases}\dfrac{at}{1+at}, & n=0,\\ \dfrac{(at)^{n-1}}{(1+at)^{n+1}}, & n=1,2,3,\ldots\end{cases}$$

where a is a constant. **Show that $\{X(t)\}$ is not stationary.**

Solution

The $n=0$ term contributes $0\cdot\frac{at}{1+at}=0$, so

$$\begin{aligned}E[X(t)]&=\sum_{n=0}^{\infty} n\,P(X(t)=n)=\sum_{n=1}^{\infty} n\,\frac{(at)^{n-1}}{(1+at)^{n+1}}\\&=\frac{1}{(1+at)^2}\sum_{n=1}^{\infty} n\left(\frac{at}{1+at}\right)^{n-1}\\&=\frac{1}{(1+at)^2}\left[1+2\left(\frac{at}{1+at}\right)+3\left(\frac{at}{1+at}\right)^2+\cdots\right].\end{aligned}$$

Let $y=\dfrac{at}{1+at}$ $(0<y<1)$. Using $1+2y+3y^2+\cdots=(1-y)^{-2}$,

$$E[X(t)]=\frac{1}{(1+at)^2}\,(1-y)^{-2}.$$

Substituting y:

$$1-y=1-\frac{at}{1+at}=\frac{1+at-at}{1+at}=\frac{1}{1+at}\;\Rightarrow\;(1-y)^{-2}=(1+at)^2.$$

$$\therefore\; E[X(t)]=\frac{1}{(1+at)^2}\,(1+at)^2=1\;\Rightarrow\;\boxed{E[X(t)]=1,\ \text{a constant.}}$$

(ii) Variance of $X(t)$
Since $E[X(t)]=1$,

$$\begin{aligned}\mathrm{Var}[X(t)]&=E[X^2(t)]-(E[X(t)])^2=E[X^2(t)]-1\\&=\sum_n n^2\,P(X(t)=n)-1\\&=\sum_n (n^2+n-n)\,P(X(t)=n)-1.\end{aligned}$$

$$\begin{aligned}\mathrm{Var}[X(t)]&=\sum_n n(n+1)\,P(X(t)=n)-\sum_n n\,P(X(t)=n)-1\\&=\sum_n n(n+1)\,P(X(t)=n)-1-1\end{aligned}$$

$$= \sum_n n(n+1)\,P(X(t)=n) - 2,$$

since $\sum_n n\,P(X(t)=n) = E[X(t)] = 1$.
Substituting the given probabilities (the $n=0$ term is zero):

$$\begin{aligned}\mathrm{Var}[X(t)] &= \sum_{n=1}^{\infty} n(n+1)\frac{(at)^{n-1}}{(1+at)^{n+1}} - 2 \\ &= \frac{1}{(1+at)^2}\left[1\cdot 2 + 2\cdot 3\,y + 3\cdot 4\,y^2 + \cdots\right] - 2 \\ &= \frac{2}{(1+at)^2}\left[1 + 3y + 6y^2 + \cdots\right] - 2, \qquad y = \frac{at}{1+at}.\end{aligned}$$

Using $1 + 3y + 6y^2 + \cdots = (1-y)^{-3}$:

$$\begin{aligned}\mathrm{Var}[X(t)] &= \frac{2}{(1+at)^2}(1-y)^{-3} - 2 = \frac{2}{(1+at)^2}\left(\frac{1}{1+at}\right)^{-3} - 2 \\ &= \frac{2}{(1+at)^2}\,(1+at)^3 - 2 = 2(1+at) - 2 = 2at.\end{aligned}$$

$$\boxed{\mathrm{Var}[X(t)] = 2at}$$

$E[X(t)]$ is constant but $\mathrm{Var}[X(t)]$ depends on t.

$$\boxed{\therefore\ \{X(t)\} \text{ is a Non-Stationary (Evolutionary) Process.}}$$

2.5 Problem 2 – Stationarity of $X(t) = \cos(\omega_0 t + \theta)$

Question

A radio station broadcasts its signal in the form of a cosine wave. Receivers at different places receive signals modelled as

$$X(t) = \cos(\omega_0 t + \theta),$$

where ω_0 is a constant and θ is a uniformly distributed random variable in $(-\pi, \pi)$. **Examine the stationarity of $\{X(t)\}$.**

Solution

The pdf of θ is $f(\theta) = \dfrac{1}{2\pi}$, $-\pi < \theta < \pi$ (and 0 otherwise).

$$\begin{aligned}E[X(t)] &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos(\omega_0 t + \theta)\,d\theta = \frac{1}{2\pi}\Big[\sin(\omega_0 t + \theta)\Big]_{-\pi}^{\pi} \\ &= \frac{1}{2\pi}\Big[\sin(\omega_0 t + \pi) - \sin(\omega_0 t - \pi)\Big] \\ &= \frac{1}{2\pi}\Big[-\sin\omega_0 t - (-\sin\omega_0 t)\Big] = 0.\end{aligned}$$

$$\boxed{E[X(t)] = 0,\ \text{a constant.}}$$

(ii) Variance of $X(t)$

$$\begin{aligned}\text{Var}[X(t)] &= E[X^2(t)] - (E[X(t)])^2 = E[X^2(t)] \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos^2(\omega_0 t + \theta)\, d\theta \\ &= \frac{1}{4\pi}\int_{-\pi}^{\pi} \left[1 + \cos(2\omega_0 t + 2\theta)\right] d\theta \\ &= \frac{1}{4\pi}\left[\theta + \frac{\sin(2\omega_0 t + 2\theta)}{2}\right]_{-\pi}^{\pi}\end{aligned}$$

$$\begin{aligned}\text{Var}[X(t)] &= \frac{1}{4\pi}\left[\pi - (-\pi) + \tfrac{1}{2}\big(\sin(2\omega_0 t + 2\pi) - \sin(2\omega_0 t - 2\pi)\big)\right] \\ &= \frac{1}{4\pi}\left[2\pi + \tfrac{1}{2}\big(\sin 2\omega_0 t - \sin 2\omega_0 t\big)\right] \\ &= \frac{1}{4\pi}\left[2\pi + 0\right] = \frac{1}{2}.\end{aligned}$$

$$\boxed{\text{Var}[X(t)] = \tfrac{1}{2},\ \text{a constant.}}$$

Conclusion
Thus both the mean and the variance are constants:

$$\boxed{X(t) = \cos(\omega_0 t + \theta) \text{ is a Stationary Random Process.}}$$

2.6 Problem 3 – Stationarity of $X(t) = A\cos(\omega_0 t + \theta)$

Question

Show that the random process

$$X(t) = A\cos(\omega_0 t + \theta)$$

is a **Stationary Random Process**, where A and ω_0 are constants and θ is a uniformly distributed random variable in the interval $(0, 2\pi)$.

Solution

The pdf of θ is $f(\theta) = \dfrac{1}{2\pi},\ 0 < \theta < 2\pi$ (and 0 otherwise).

$$\begin{aligned}E[X(t)] &= \int_0^{2\pi} A\cos(\omega_0 t + \theta)\,\frac{1}{2\pi}\, d\theta = \frac{A}{2\pi}\Big[\sin(\omega_0 t + \theta)\Big]_0^{2\pi} \\ &= \frac{A}{2\pi}\Big[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t + 0)\Big] \\ &= \frac{A}{2\pi}\Big[\sin\omega_0 t - \sin\omega_0 t\Big] = 0.\end{aligned}$$

$$\boxed{E[X(t)] = 0,\ \text{a constant (not a function of } t).}$$

(ii) Variance of $X(t)$

$$\begin{aligned}\text{Var}[X(t)] &= E[X^2(t)] - (E[X(t)])^2 = E[X^2(t)] \\ &= \frac{A^2}{2\pi}\int_0^{2\pi} \cos^2(\omega_0 t + \theta)\, d\theta \\ &= \frac{A^2}{2\pi}\int_0^{2\pi} \frac{1+\cos(2\omega_0 t + 2\theta)}{2}\, d\theta \\ &= \frac{A^2}{4\pi}\left[\theta + \frac{\sin(2\omega_0 t + 2\theta)}{2}\right]_0^{2\pi}\end{aligned}$$

$$\begin{aligned}\text{Var}[X(t)] &= \frac{A^2}{4\pi}\left[(2\pi - 0) + \tfrac{1}{2}\big(\sin(2\omega_0 t + 4\pi) - \sin(2\omega_0 t + 0)\big)\right] \\ &= \frac{A^2}{4\pi}\left[2\pi + \tfrac{1}{2}\big(\sin 2\omega_0 t - \sin 2\omega_0 t\big)\right] \\ &= \frac{A^2}{4\pi}\left[2\pi + 0\right] = \frac{A^2}{2}.\end{aligned}$$

$$\boxed{\text{Var}[X(t)] = \frac{A^2}{2},\ \text{a constant.}}$$

Conclusion
Since both $E[X(t)]$ and $\text{Var}[X(t)]$ are constants (independent of t),

$$\boxed{\therefore\ X(t) = A\cos(\omega_0 t + \theta) \text{ is a Stationary Random Process.}}$$

2.7 Example – Stationarity of $X(t) = A\sin(\omega t + \phi)$

Question

Check the stationarity of the collection of signals given by

$$X(t) = A\sin(\omega t + \phi),$$

where A and ω are constants, and ϕ is a uniformly distributed random variable in $(-\pi, \pi)$.

Solution

The pdf of ϕ is $f(\phi) = \dfrac{1}{2\pi},\ -\pi < \phi < \pi$ (and 0 otherwise).

$$\begin{aligned}E[X(t)] &= \frac{A}{2\pi}\int_{-\pi}^{\pi} \sin(\omega t + \phi)\, d\phi = \frac{A}{2\pi}\Big[-\cos(\omega t + \phi)\Big]_{-\pi}^{\pi} \\ &= \frac{A}{2\pi}\Big[-\cos(\omega t + \pi) + \cos(\omega t - \pi)\Big] \\ &= \frac{A}{2\pi}\Big[\cos\omega t - \cos\omega t\Big] = 0.\end{aligned}$$

$$\boxed{E[X(t)] = 0,\ \text{a constant.}}$$

(ii) Variance of $X(t)$

Since $E[X(t)] = 0$,

$$\begin{aligned}\text{Var}[X(t)] = E[X^2(t)] &= \frac{A^2}{2\pi}\int_{-\pi}^{\pi} \sin^2(\omega t + \phi)\, d\phi \\ &= \frac{A^2}{2\pi}\int_{-\pi}^{\pi} \frac{1-\cos(2\omega t + 2\phi)}{2}\, d\phi \\ &= \frac{A^2}{4\pi}\left[\phi - \frac{\sin(2\omega t + 2\phi)}{2}\right]_{-\pi}^{\pi}\end{aligned}$$

$$\begin{aligned}\text{Var}[X(t)] &= \frac{A^2}{4\pi}\left[(\pi - (-\pi)) - \tfrac{1}{2}\big(\sin(2\omega t + 2\pi) - \sin(2\omega t - 2\pi)\big)\right] \\ &= \frac{A^2}{4\pi}\left[2\pi - \tfrac{1}{2}\big(\sin 2\omega t - \sin 2\omega t\big)\right] \\ &= \frac{A^2}{4\pi}\left[2\pi - 0\right] = \frac{A^2}{2}.\end{aligned}$$

$$\boxed{\text{Var}[X(t)] = \frac{A^2}{2},\ \text{a constant.}}$$

Conclusion
Since both $E[X(t)]$ and $\text{Var}[X(t)]$ are constants (independent of t),

$$\boxed{\therefore\ X(t) = A\sin(\omega t + \phi) \text{ is a Stationary Random Process.}}$$

Recall: WSS Process

$$(i)\ E[X(t)] = \text{constant}, \qquad (ii)\ R_X(t_1, t_2) = R_X(t_1 - t_2) = R_X(\tau).$$

2.8 Problem – $X(t) = A\cos(\omega_0 t + \theta)$ is a WSS Process

Question

Prove that the random process

$$X(t) = A\cos(\omega_0 t + \theta)$$

is a **Wide-Sense Stationary (WSS)** process, where A and ω_0 are constants and θ is a uniformly distributed random variable in $(0, 2\pi)$.

Solution

The pdf of θ is $f(\theta) = \dfrac{1}{2\pi},\ 0 < \theta < 2\pi$ (and 0 otherwise).

$$\begin{aligned}E[X(t)] = \int_0^{2\pi} A\cos(\omega_0 t + \theta)\,\frac{1}{2\pi}\, d\theta &= \frac{A}{2\pi}\Big[\sin(\omega_0 t + \theta)\Big]_0^{2\pi} \\ &= \frac{A}{2\pi}\Big[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t + 0)\Big] \\ &= \frac{A}{2\pi}\Big[\sin\omega_0 t - \sin\omega_0 t\Big] = 0.\end{aligned}$$

$$\boxed{E[X(t)] = 0, \text{ a constant.}}$$

(ii) ACF of $X(t)$
Using $\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$:

$$\begin{aligned} R_X(t_1, t_2) &= E[X(t_1)X(t_2)] \\ &= A^2\, E\big[\cos(\omega_0 t_1 + \theta)\cos(\omega_0 t_2 + \theta)\big] \\ &= \frac{A^2}{2} E\big[\cos(\omega_0(t_1+t_2) + 2\theta) + \cos(\omega_0(t_1 - t_2))\big] \\ &= \frac{A^2}{2}\Big[\frac{1}{2\pi}\int_0^{2\pi} \cos(\omega_0(t_1+t_2) + 2\theta)\, d\theta \\ &\qquad + \cos(\omega_0(t_1 - t_2))\Big]. \end{aligned}$$

$$\begin{aligned} \int_0^{2\pi} \cos(\omega_0(t_1+t_2) + 2\theta)\, d\theta &= \frac{1}{2}\Big[\sin(\omega_0(t_1+t_2) + 2\theta)\Big]_0^{2\pi} \\ &= \frac{1}{2}\Big[\sin(\omega_0(t_1+t_2) + 4\pi) - \sin(\omega_0(t_1+t_2) + 0)\Big] \\ &= \frac{1}{2}\Big[\sin\omega_0(t_1+t_2) - \sin\omega_0(t_1+t_2)\Big] = 0. \end{aligned}$$

$$\therefore\ R_X(t_1, t_2) = \frac{A^2}{2}\,[0 + \cos(\omega_0(t_1 - t_2))] = \frac{A^2}{2}\cos[\omega_0(t_1 - t_2)].$$

Conclusion
With $\tau = t_1 - t_2$,

$$\boxed{R_X(t_1, t_2) = \frac{A^2}{2}\cos[\omega_0(t_1 - t_2)] = \frac{A^2}{2}\cos\omega_0\tau = R_X(\tau).}$$

Hence $R_X(t_1, t_2)$ is a function of the time difference only.

$$\boxed{\begin{aligned} E[X(t)] &= \text{constant}, \\ R_X(t_1, t_2) &= R_X(t_1 - t_2). \end{aligned}}$$

$$\therefore\ X(t) = A\cos(\omega_0 t + \theta) \text{ is a } \textbf{WSS Process}.$$

2.9 Problem – $Y(t) = X(t)\cos(\omega t + \phi)$ is a WSS Process

Question

Given the random process

$$Y(t) = X(t)\cos(\omega t + \phi),$$

where $X(t)$ is a WSS process, ω is a constant, and ϕ is a uniformly distributed random variable in $(-\pi, \pi)$, independent of $X(t)$, prove that $Y(t)$ is also a WSS process.

Solution

Since $X(t)$ is WSS:

$$(i) \quad E[X(t)] = \text{constant} = k,$$
$$(ii) \quad R_X(t_1, t_2) = R_X(t_1 - t_2) = h(t_1 - t_2).$$

Also, ϕ is uniformly distributed in $(-\pi, \pi)$ and is independent of $X(t)$:

$$f(\phi) = \begin{cases} \frac{1}{2\pi}, & -\pi < \phi < \pi, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Mean of $Y(t)$
Since $X(t)$ and ϕ are independent,

$$\begin{aligned} E[Y(t)] &= E[X(t)\cos(\omega t+\phi)] = E[X(t)]\,E[\cos(\omega t+\phi)] \\ &= \frac{k}{2\pi}\Big[\sin(\omega t+\phi)\Big]_{-\pi}^{\pi} \\ &= \frac{k}{2\pi}\Big[\sin(\omega t+\pi)-\sin(\omega t-\pi)\Big] \\ &= \frac{k}{2\pi}\Big[-\sin\omega t+\sin\omega t\Big] = 0. \end{aligned}$$

$$\boxed{E[Y(t)] = 0,\ \text{a constant.}}$$

(b) ACF of $Y(t)$
Since $X(t)$ and ϕ are independent,

$$\begin{aligned} R_Y(t_1,t_2) &= E[Y(t_1)Y(t_2)] \\ &= E\big[X(t_1)\cos(\omega t_1+\phi)\,X(t_2)\cos(\omega t_2+\phi)\big] \\ &= E[X(t_1)X(t_2)]\;E\big[\cos(\omega t_1+\phi)\cos(\omega t_2+\phi)\big] \\ &= R_X(t_1,t_2)\;E\big[\cos(\omega t_1+\phi)\cos(\omega t_2+\phi)\big]. \end{aligned}$$

Using $\cos A\cos B = \frac{1}{2}[\cos(A+B)+\cos(A-B)]$:

$$\begin{aligned} &E[\cos(\omega t_1+\phi)\cos(\omega t_2+\phi)] \\ &= \frac{1}{2}E\big[\cos(\omega(t_1+t_2)+2\phi)+\cos(\omega(t_1-t_2))\big] \\ &= \frac{1}{2}\Big[\frac{1}{2\pi}\int_{-\pi}^{\pi}\cos(\omega(t_1+t_2)+2\phi)\,d\phi \\ &\qquad +\cos(\omega(t_1-t_2))\Big]. \end{aligned}$$

$$\begin{aligned} \int_{-\pi}^{\pi}\cos(\omega(t_1+t_2)+2\phi)\,d\phi &= \frac{1}{2}\Big[\sin(\omega(t_1+t_2)+2\phi)\Big]_{-\pi}^{\pi} \\ &= \frac{1}{2}\Big[\sin(\omega(t_1+t_2)+2\pi)-\sin(\omega(t_1+t_2)-2\pi)\Big] \\ &= \frac{1}{2}\Big[\sin\omega(t_1+t_2)-\sin\omega(t_1+t_2)\Big] = 0. \end{aligned}$$

$$\therefore\ E[\cos(\omega t_1+\phi)\cos(\omega t_2+\phi)] = \frac{1}{2}\cos[\omega(t_1-t_2)].$$

Conclusion
Substituting back,

$$R_Y(t_1,t_2) = R_X(t_1,t_2)\cdot\frac{1}{2}\cos[\omega(t_1-t_2)] = \frac{1}{2}R_X(t_1-t_2)\cos[\omega(t_1-t_2)].$$

$$\boxed{R_Y(t_1,t_2) = \tfrac{1}{2}R_X(\tau)\cos\omega\tau = R_Y(\tau),\quad \tau = t_1-t_2.}$$

$$\boxed{\begin{array}{ll}(i) & E[Y(t)] = \text{constant},\\ (ii) & R_Y(t_1,t_2) = R_Y(t_1-t_2).\end{array}}$$

$\therefore\ Y(t)$ is a **WSS Process**.

2.10 Exercise

Exercise Problem

Question: Given the random process $X(t) = A\cos(\omega_0 t + \theta)$, where A and ω_0 are constants and θ is uniformly distributed in $(0, \pi)$, **check the stationarity of the random process.**
Answer: Here $f(\theta) = \frac{1}{\pi}$, $0 < \theta < \pi$, and

$$E[X(t)] = \frac{A}{\pi}\Big[\sin(\omega_0 t + \theta)\Big]_0^{\pi} = \frac{A}{\pi}\Big[\sin(\omega_0 t + \pi) - \sin(\omega_0 t + 0)\Big]$$
$$= \frac{A}{\pi}\Big[-\sin\omega_0 t - \sin\omega_0 t\Big] = -\frac{2A}{\pi}\sin\omega_0 t,$$

which depends on t. Hence $X(t)$ **is not stationary.**

2.11 Problem – $X(t) = A\cos\omega t + B\sin\omega t$ is a WSS Process

Question

Show that the random process

$$X(t) = A\cos\omega t + B\sin\omega t$$

is a WSS process, where A and B are independent random variables with mean zero and variance 1.

Solution

A and B are independent with mean 0 and variance 1:

$$E(A) = 0, \qquad E(B) = 0. \tag{1}$$

$$\mathrm{Var}(A) = 1 = E(A^2) - (E(A))^2,$$
$$\mathrm{Var}(B) = 1 = E(B^2) - (E(B))^2.$$

Using (1): $E(A^2) = 1, \quad E(B^2) = 1.$ (2)
Since A and B are independent: $E(AB) = E(A)E(B) = 0 = E(BA).$ (3)

Mean and ACF
Mean: $E[X(t)] = E(A)\cos\omega t + E(B)\sin\omega t = 0$ (constant).
ACF:

$$R_X(t_1, t_2) = E\big[(A\cos\omega t_1 + B\sin\omega t_1)(A\cos\omega t_2 + B\sin\omega t_2)\big]$$
$$= E(A^2)\cos\omega t_1\cos\omega t_2 + E(AB)\cos\omega t_1\sin\omega t_2$$
$$+ E(BA)\sin\omega t_1\cos\omega t_2 + E(B^2)\sin\omega t_1\sin\omega t_2$$
$$= \cos\omega t_1\cos\omega t_2 + \sin\omega t_1\sin\omega t_2 \quad \text{(by (2), (3))}$$
$$= \cos[\omega(t_1 - t_2)].$$

Conclusion

$$\boxed{E[X(t)] = 0 \text{ (constant)}}$$

$$\boxed{R_X(t_1, t_2) = \cos[\omega(t_1 - t_2)] = \cos \omega\tau = R_X(\tau)}$$

$$\boxed{\begin{array}{ll}(i) & E[X(t)] = \text{constant},\\ (ii) & R_X(t_1, t_2) = \text{a function of time difference.}\end{array}}$$

$$\boxed{\text{Therefore, } X(t) = A\cos\omega t + B\sin\omega t \text{ is WSS.}}$$

2.12 Problem – $X(t) = Y\cos\omega t + Z\sin\omega t$ is a WSS Process

Question

Show that the random process

$$X(t) = Y\cos\omega t + Z\sin\omega t$$

is a WSS process, where Y and Z are independent random variables which assume the values -2 and 1 with probabilities $\frac{1}{3}$ and $\frac{2}{3}$ respectively.

Solution

y	-2	1
$P(Y=y)$	$\frac{1}{3}$	$\frac{2}{3}$

z	-2	1
$P(Z=z)$	$\frac{1}{3}$	$\frac{2}{3}$

$$E(Y) = (-2)\left(\tfrac{1}{3}\right) + (1)\left(\tfrac{2}{3}\right) = 0; \quad \text{similarly } E(Z) = 0. \tag{1}$$

$$E(Y^2) = (-2)^2\left(\tfrac{1}{3}\right) + 1^2\left(\tfrac{2}{3}\right) = \tfrac{4}{3} + \tfrac{2}{3} = 2; \quad \text{sim. } E(Z^2) = 2. \tag{2}$$

$$E(YZ) = E(Y)E(Z) = 0 = E(ZY) \quad \text{(independence)}. \tag{3}$$

Mean and ACF

Mean: $E[X(t)] = E(Y)\cos\omega t + E(Z)\sin\omega t = 0$ (by (1)).

ACF:

$$\begin{aligned} R_X(t_1, t_2) &= E\big[(Y\cos\omega t_1 + Z\sin\omega t_1)(Y\cos\omega t_2 + Z\sin\omega t_2)\big] \\ &= E(Y^2)\cos\omega t_1\cos\omega t_2 + E(YZ)\cos\omega t_1\sin\omega t_2 \\ &\quad + E(ZY)\sin\omega t_1\cos\omega t_2 + E(Z^2)\sin\omega t_1\sin\omega t_2 \\ &= 2\cos\omega t_1\cos\omega t_2 + 0 + 0 + 2\sin\omega t_1\sin\omega t_2 \\ &= 2\cos[\omega(t_1 - t_2)] \qquad \text{(by (2), (3))}. \end{aligned}$$

Conclusion

$$\boxed{E[X(t)] = 0, \text{ constant}} \qquad \boxed{R_X(t_1, t_2) = 2\cos[\omega(t_1 - t_2)] = R_X(\tau)}$$

Since

1. $E[X(t)]$ is a constant, and
2. $R_X(t_1, t_2)$ depends only on the time difference $t_1 - t_2$,

$$\boxed{\therefore\ X(t) \text{ is a WSS Process.}}$$

2.13 Exercise

Exercise Problem

Consider the random process

$$X(t) = A\cos\omega t + B\sin\omega t,$$

where A and B are uncorrelated random variables with zero mean and the same variance σ^2. **Prove that $X(t)$ is a WSS process.**

Hint: $E(AB) = E(A)E(B) = 0$ (uncorrelated), $E(A^2) = E(B^2) = \sigma^2$; show that $E[X(t)] = 0$ and $R_X(\tau) = \sigma^2\cos\omega\tau$.

2.14 Definitions

Cross-Correlation Function (CCF)

The **CCF** of two random processes $\{X(t)\}$ and $\{Y(t)\}$ is $R_{XY}(t_1, t_2) = E\big[X(t_1)Y(t_2)\big]$. If the processes are jointly stationary, it is a function of the time difference only.

Jointly WSS Processes

$\{X(t)\}$ and $\{Y(t)\}$ are **jointly WSS** if (1) each is individually WSS, and (2) $R_{XY}(t_1, t_2)$ is a function of the time difference only.

2.15 Autocorrelation Function (ACF) – Definition

Definition

Let $\{x(t)\}$ be a wide-sense stationary random process. Its autocorrelation function (ACF) depends only on the time difference $\tau = t_1 - t_2$:

$$\begin{aligned} R(\tau) &= E[x(t_1)x(t_2)], \qquad \tau = t_1 - t_2, \\ &= E[x(t)\,x(t-\tau)]. \end{aligned}$$

Thus $R(\tau)$ indicates how similar two signal values are when separated by τ units of time.

2.16 Properties of the ACF – Property 1

Property 1

$R(\tau)$ is an even function of τ: $\quad R(\tau) = R(-\tau)$.

Proof

By definition, $R(\tau) = E[x(t)x(t-\tau)]$, so $R(-\tau) = E[x(t)x(t+\tau)]$.
Let $\lambda = t + \tau \;\Rightarrow\; t = \lambda - \tau$:

$$R(-\tau) = E[x(\lambda-\tau)\,x(\lambda)] = E[x(\lambda)\,x(\lambda-\tau)]$$

$$= R(\tau) \qquad \text{(WSS: independent of } \lambda\text{)}.$$

Hence $R(\tau)$ is even.

2.17 Properties of the ACF – Property 2

Property 2

The ACF is bounded by its zero-lag value: $|R(\tau)| \leq R(0)$.

Proof

$R(\tau) = E[x(t_1)x(t_2)]$ with $\tau = t_1 - t_2$, and $R(0) = E[x^2(t)]$ for every t.
Consider $E\left[(x(t_1) + x(t_2))^2\right] \geq 0$. Expanding and using stationarity,

$$E[x^2(t_1)] + E[x^2(t_2)] + 2E[x(t_1)x(t_2)] \geq 0$$
$$2R(0) + 2R(\tau) \geq 0 \;\Rightarrow\; R(\tau) \geq -R(0). \tag{1}$$

Similarly, from $E\left[(x(t_1) - x(t_2))^2\right] \geq 0$,

$$E[x^2(t_1)] + E[x^2(t_2)] - 2E[x(t_1)x(t_2)] \geq 0$$
$$2R(0) - 2R(\tau) \geq 0 \;\Rightarrow\; R(\tau) \leq R(0). \tag{2}$$

Combining (1) and (2):

$$-R(0) \leq R(\tau) \leq R(0) \quad \Longrightarrow \quad \boxed{|R(\tau)| \leq R(0).}$$

2.18 Properties of the ACF – Property 3

Property 3

If $x(t)$ has mean μ_x (and no periodic component), then $\lim_{\tau\to\infty} R(\tau) = \mu_x^2$.

Proof

$R(\tau) = E[x(t)x(t-\tau)]$. As $\tau \to \infty$, the values $x(t)$ and $x(t-\tau)$ become statistically independent, so

$$\lim_{\tau\to\infty} R(\tau) = E[x(t)] \cdot E[x(t-\tau)] = \mu_x \cdot \mu_x = \mu_x^2.$$

2.19 Problem 1 – Suitable ACF

Problem 1

Which one of the following is a suitable ACF?
(i) $A\sin(\omega\tau)$ (ii) $A\cos(\omega\tau)$

Solution

An ACF $R(\tau)$ must be an even function of τ. Cosine is even, while sine is odd, so $A\sin(\omega\tau)$ is not an ACF.
For $R(\tau) = A\cos(\omega\tau)$ with $A > 0$: $R(0) = A\cos 0 = A$ and

$$|R(\tau)| = A|\cos(\omega\tau)| \le A = R(0),$$

which satisfies the ACF property. $\therefore$ $A\cos(\omega\tau)$ is a valid ACF.

2.20 Problem 2 – Mean and Variance from ACF

Problem 2

Given the ACF of a stationary R.P. $\{x(t)\}$ as $R(\tau) = 25 + \dfrac{4}{1+6\tau^2}$, find the mean and variance of the process.

Solution

Using $\lim_{\tau\to\infty} R(\tau) = \mu_x^2$:

$$\mu_x^2 = \lim_{\tau\to\infty}\left(25 + \frac{4}{1+6\tau^2}\right) = 25 + \frac{4}{1+6(\infty)^2}$$
$$= 25 + 0 = 25 \;\Rightarrow\; |\mu_x| = 5.$$

So $\mu_x = \pm 5$ (the ACF fixes only μ_x^2; we take $\mu_x = 5$).
Variance: $\mathrm{Var}(x(t)) = E[x^2(t)] - \mu_x^2$, where

$$E[x^2(t)] = R(0) = 25 + \frac{4}{1+6(0)^2} = 25 + 4 = 29.$$

$$\mathrm{Var}(x(t)) = 29 - 25 = 4.$$

$$\therefore \quad \boxed{\mu_x = 5 \text{ (or } -5), \qquad \sigma_x^2 = 4.}$$

2.21 Problem 3 – Mean, Variance and Covariance

Problem 3

Let $x(t)$ be a stationary R.P. with mean 3 and ACF $R_x(t_1, t_2) = 9 + 4e^{-0.2|t_1 - t_2|}$. Determine the mean, variance, and covariance of the random variables $X(5)$ and $X(8)$.

Solution

Since the process is stationary,

$$R(\tau) = 9 + 4e^{-0.2|\tau|}, \quad \tau = t_1 - t_2, \qquad E[x(t)] = \mu_x = 3 \ \ \forall t.$$

Let $Z = X(5)$ and $W = X(8)$. Then $E[Z] = E[X(5)] = 3$ and $E[W] = E[X(8)] = 3$.
Variance:

$$E[Z^2] = R(0) = 9 + 4e^{-0.2(0)} = 9 + 4 = 13,$$

$$\mathrm{Var}(Z) = E[Z^2] - (E[Z])^2 = 13 - 3^2 = 13 - 9 = 4.$$

Similarly, $\mathrm{Var}(W) = 4$.
Covariance: Since $|5 - 8| = 3$,

$$E[X(5)X(8)] = R(3) = 9 + 4e^{-0.2\times 3} = 9 + 4e^{-0.6}.$$

$$\begin{aligned}\mathrm{Cov}(Z, W) &= E[ZW] - E[Z]E[W] \\ &= (9 + 4e^{-0.6}) - 3 \cdot 3 = 4e^{-0.6} \approx 2.195.\end{aligned}$$

$$\boxed{\mu_x = 3, \quad \mathrm{Var} = 4, \quad \mathrm{Cov}(X(5), X(8)) = 4e^{-0.6}}$$

2.22 Problem 4 – Second Moment of $X(8) - X(5)$

Problem 4

For a stationary random process, the ACF is $R(\tau) = Ae^{-|\tau|}$, where A is a constant. Find the second moment of the random variable $X(8) - X(5)$.

Solution

$R(\tau) = Ae^{-|\tau|} \ \Rightarrow \ R(t_1, t_2) = Ae^{-|t_1 - t_2|}$, and $R(0) = Ae^{-0} = A$.

$$\begin{aligned}E\big[(X(8) - X(5))^2\big] &= E[X^2(8)] + E[X^2(5)] - 2E[X(8)X(5)] \\ &= R(0) + R(0) - 2R(8, 5) \\ &= 2A - 2Ae^{-|8-5|} = 2A - 2Ae^{-3} \\ &= 2A(1 - e^{-3}).\end{aligned}$$

$$\boxed{E\big[(X(8) - X(5))^2\big] = 2A(1 - e^{-3})}$$

2.23 Cross-Correlation Function (CCF) – Property 1

Definition: CCF

For jointly stationary random processes $x(t)$ and $y(t)$, the CCF is a function of the time difference: $R_{xy}(\tau) = E\big[x(t)\, y(t - \tau)\big]$.

Property 1

$R_{xy}(-\tau) = R_{yx}(\tau)$.

Proof

$R_{xy}(-\tau) = E[x(t)\,y(t+\tau)]$. Let $\lambda = t+\tau \Rightarrow t = \lambda - \tau$:

$$R_{xy}(-\tau) = E[x(\lambda-\tau)\,y(\lambda)] = E[y(\lambda)\,x(\lambda-\tau)] = R_{yx}(\tau).$$

2.24 CCF – Property 2

Property 2

For any two jointly WSS random processes: $|R_{xy}(\tau)| \le \frac{1}{2}\big(R_{xx}(0) + R_{yy}(0)\big)$.

Proof

We know $R_{xx}(0) = E[x^2(t)]$, $R_{yy}(0) = E[y^2(t)]$. (1)

Consider $E\big(x(t_1) + y(t_2)\big)^2 \ge 0$, with $\tau = t_1 - t_2$.
Expanding and using (1):

$$\begin{aligned} E[x^2(t_1)] + E[y^2(t_2)] + 2E[x(t_1)y(t_2)] &\ge 0 \\ R_{xx}(0) + R_{yy}(0) + 2R_{xy}(\tau) &\ge 0 \\ \Rightarrow\; R_{xy}(\tau) &\ge -\tfrac{1}{2}\big(R_{xx}(0) + R_{yy}(0)\big). \end{aligned} \quad (2)$$

Now consider $E\big(x(t_1) - y(t_2)\big)^2 \ge 0$:

$$\begin{aligned} E[x^2(t_1)] + E[y^2(t_2)] - 2E[x(t_1)y(t_2)] &\ge 0 \\ R_{xx}(0) + R_{yy}(0) - 2R_{xy}(\tau) &\ge 0 \\ \Rightarrow\; R_{xy}(\tau) &\le \tfrac{1}{2}\big(R_{xx}(0) + R_{yy}(0)\big). \end{aligned} \quad (3)$$

Combining (2) and (3):

$$\boxed{|R_{xy}(\tau)| \le \tfrac{1}{2}\big(R_{xx}(0) + R_{yy}(0)\big)}$$

2.25 CCF – Property 3

Property 3

$|R_{xy}(\tau)| \le \sqrt{R_{xx}(0)\,R_{yy}(0)}$.

Proof

Consider

$$E\left(\frac{x(t_1)}{\sqrt{R_{xx}(0)}} + \frac{y(t_2)}{\sqrt{R_{yy}(0)}}\right)^2 \ge 0.$$

Expanding,

$$\frac{R_{xx}(0)}{R_{xx}(0)} + \frac{R_{yy}(0)}{R_{yy}(0)} + \frac{2R_{xy}(\tau)}{\sqrt{R_{xx}(0)R_{yy}(0)}} \geq 0$$

$$\Rightarrow \; 2 + \frac{2R_{xy}(\tau)}{\sqrt{R_{xx}(0)R_{yy}(0)}} \geq 0$$

$$\Rightarrow \; R_{xy}(\tau) \geq -\sqrt{R_{xx}(0)R_{yy}(0)}. \tag{2}$$

Similarly, from

$$E\left(\frac{x(t_1)}{\sqrt{R_{xx}(0)}} - \frac{y(t_2)}{\sqrt{R_{yy}(0)}}\right)^2 \geq 0,$$

$$2 - \frac{2R_{xy}(\tau)}{\sqrt{R_{xx}(0)R_{yy}(0)}} \geq 0 \;\Rightarrow\; R_{xy}(\tau) \leq \sqrt{R_{xx}(0)R_{yy}(0)}. \tag{3}$$

Combining (2) and (3):

$$-\sqrt{R_{xx}(0)R_{yy}(0)} \leq R_{xy}(\tau) \leq \sqrt{R_{xx}(0)R_{yy}(0)}$$

$$\Longrightarrow \boxed{|R_{xy}(\tau)| \leq \sqrt{R_{xx}(0)R_{yy}(0)}}$$

2.26 CCF – Property 4

Property 4

If $\{x(t)\}$ and $\{y(t)\}$ are two independent (WSS) random processes, then $R_{xy}(\tau) = \mu_x\mu_y$, where $\mu_x = E[x(t)]$, $\mu_y = E[y(t)]$.

Proof

$R_{xy}(\tau) = E\big[x(t)\,y(t-\tau)\big]$. Since the processes are independent, $x(t)$ and $y(t-\tau)$ are independent for all t, τ; hence

$$R_{xy}(\tau) = E[x(t)]\,E[y(t-\tau)] = \mu_x\mu_y,$$

because the means of WSS processes do not depend on t.

2.27 Problem – CCF of $W(t)$ and $Z(t)$

Problem

Given two stationary random processes $W(t) = A(t) + B(t)$ and $Z(t) = A(t) - B(t)$, where $A(t)$ and $B(t)$ are independent random processes with mean zero and ACFs $R_{AA}(\tau) = e^{-|\tau|}$, $R_{BB}(\tau) = 3e^{-|\tau|}$, find the cross-correlation function of $W(t)$ and $Z(t)$.

Solution

Given $E[A(t)] = 0$, $E[B(t)] = 0$.

$$\begin{aligned} R_{WZ}(\tau) &= E\big[W(t)\,Z(t-\tau)\big] \\ &= E\big[(A(t)+B(t))\,(A(t-\tau)-B(t-\tau))\big]. \end{aligned}$$

$$\begin{aligned} R_{WZ}(\tau) = E[A(t)A(t-\tau)] - E[A(t)B(t-\tau)] \\ + E[B(t)A(t-\tau)] - E[B(t)B(t-\tau)]. \end{aligned}$$

By independence and zero means, $E[A(t)B(t-\tau)] = E[A(t)]E[B(t-\tau)] = 0$ and $E[B(t)A(t-\tau)] = 0$. Hence

$$R_{WZ}(\tau) = R_{AA}(\tau) - R_{BB}(\tau) = e^{-|\tau|} - 3e^{-|\tau|} = -2e^{-|\tau|}.$$

$$\boxed{R_{WZ}(\tau) = -2e^{-|\tau|}}$$

2.28 Problem – CCF of Two Sinusoids

Problem

Two jointly WSS random processes $X(t)$ and $Y(t)$ have sample functions

$$X(t) = A_1\cos(\omega_0 t + \Theta), \qquad Y(t) = A_2\cos(\omega_0 t + \Theta + \phi),$$

where Θ is uniformly distributed in $(0, 2\pi)$ and A_1, A_2, ω_0, ϕ are constants. Find the cross-correlation function $R_{xy}(\tau)$.

Solution

The pdf of Θ is $f_\Theta(\theta) = \dfrac{1}{2\pi}$, $0 \le \theta \le 2\pi$.

$$\begin{aligned} R_{xy}(\tau) &= E\big[X(t)\,Y(t-\tau)\big] \\ &= E\big[A_1\cos(\omega_0 t + \Theta)\;A_2\cos\big(\omega_0(t-\tau) + \Theta + \phi\big)\big] \\ &= A_1A_2\,E\big[\cos(\omega_0 t + \Theta)\cos(\omega_0 t - \omega_0\tau + \Theta + \phi)\big]. \end{aligned}$$

Use $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha+\beta) + \cos(\alpha-\beta)]$ with

$$\alpha = \omega_0 t + \Theta, \qquad \beta = \omega_0 t - \omega_0\tau + \Theta + \phi.$$

$$\begin{aligned} \alpha + \beta &= (\omega_0 t + \Theta) + (\omega_0 t - \omega_0\tau + \Theta + \phi) = 2\omega_0 t - \omega_0\tau + 2\Theta + \phi, \\ \alpha - \beta &= (\omega_0 t + \Theta) - (\omega_0 t - \omega_0\tau + \Theta + \phi) = \omega_0\tau - \phi. \end{aligned}$$

Let $\psi = 2\omega_0 t - \omega_0\tau + \phi$, so that $\alpha + \beta = \psi + 2\Theta$. Then

$$\begin{aligned} R_{xy}(\tau) &= \frac{A_1A_2}{2}\,E\big[\cos(\psi + 2\Theta) + \cos(\omega_0\tau - \phi)\big] \\ &= \frac{A_1A_2}{2}\Big(E[\cos(\psi + 2\Theta)] + \cos(\omega_0\tau - \phi)\Big), \end{aligned}$$

since $\cos(\omega_0\tau - \phi)$ does not depend on Θ.

$$\begin{aligned} E[\cos(\psi + 2\Theta)] &= \frac{1}{2\pi}\int_0^{2\pi} \cos(\psi + 2\theta)\, d\theta = \frac{1}{2\pi}\left[\frac{\sin(\psi + 2\theta)}{2}\right]_0^{2\pi} \\ &= \frac{1}{4\pi}\big(\sin(\psi + 4\pi) - \sin(\psi + 0)\big) \\ &= \frac{1}{4\pi}\big(\sin\psi - \sin\psi\big) = 0. \end{aligned}$$

Therefore

$$R_{xy}(\tau) = \frac{A_1A_2}{2}\big(0 + \cos(\omega_0\tau - \phi)\big) \;\Rightarrow\; \boxed{R_{xy}(\tau) = \frac{A_1A_2}{2}\cos(\omega_0\tau - \phi)}$$

3 Power Spectral Density

PSD, its Properties, the Wiener–Khinchine Theorem and Problems

3.1 Introduction

The **power spectral density (PSD)** function is used to determine the *distribution of the power* of a random process $\{X(t)\}$ in the *frequency domain*. The autocorrelation function (ACF) $R(\tau) = E[X(t)X(t+\tau)]$ describes a wide-sense stationary (WSS) process in the time domain; the PSD carries exactly the same information, viewed in the frequency domain.

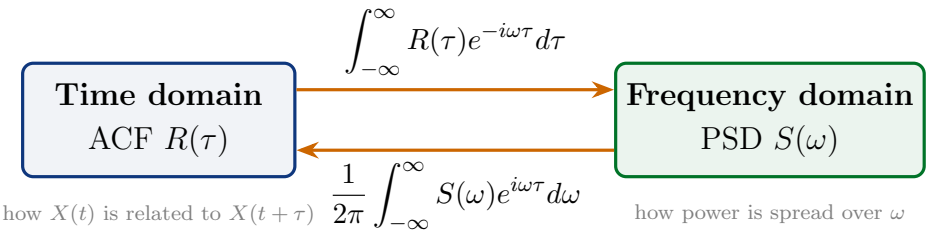


Figure 20: The ACF and the PSD form a Fourier-transform pair (Wiener–Khinchin relation).

3.2 Definition

Definition (Power Spectral Density)

The power spectral density function of a WSS process $\{X(t)\}$ is the Fourier transform of its ACF. It is denoted by $S(\omega)$ or $S_{XX}(\omega)$:

$$\boxed{S(\omega) = \int_{-\infty}^{\infty} R(\tau)\, e^{-i\omega\tau}\, d\tau}$$

If $\omega = 2\pi f$, we get the distribution of power of the random process in terms of the frequency f.

Similarly, by the inverse Fourier transform,

Inverse relation

$$R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\, e^{i\omega\tau}\, d\omega.$$

Average power

Putting $\tau = 0$ in the inverse relation,

Average power of $\{X(t)\}$

$$R(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\, d\omega = E\big[X^2(t)\big].$$

Thus $R(0)$ represents the **average power** of the random process $\{X(t)\}$.

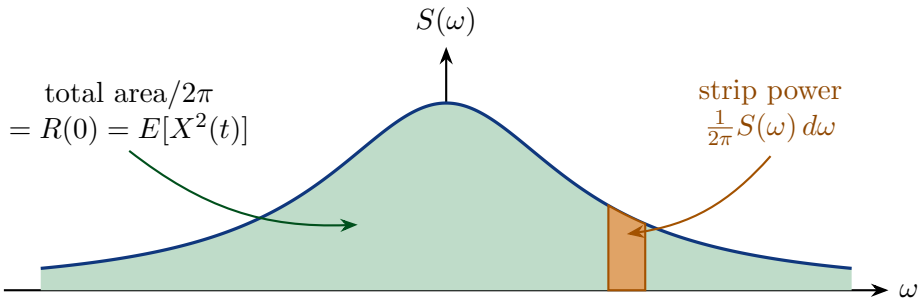


Figure 21: The total area under $S(\omega)$, divided by 2π, is the average power $R(0)$. A thin strip of width $d\omega$ carries power $\frac{1}{2\pi}S(\omega)\,d\omega$.

3.3 Properties of the PSD

Property 1: $S(\omega)$ is an even function of ω.

Proof. By definition, $S(\omega) = \int_{-\infty}^{\infty} R(\tau)e^{-i\omega\tau}\,d\tau$. Hence

$$S(-\omega) = \int_{-\infty}^{\infty} R(\tau)e^{i\omega\tau}\,d\tau.$$

Let $\tau = -u$, so $d\tau = -du$. When τ varies from $-\infty$ to ∞, u varies from ∞ to $-\infty$. Then

$$\begin{aligned} S(-\omega) &= \int_{\infty}^{-\infty} R(-u)\,e^{i\omega(-u)}\,(-du) = \int_{-\infty}^{\infty} R(-u)\,e^{-i\omega u}\,du \\ &= \int_{-\infty}^{\infty} R(u)\,e^{-i\omega u}\,du \qquad (\because R(\tau) \text{ is even} \Rightarrow R(-u) = R(u)) \\ &= S(\omega) \qquad \text{(by definition)}. \end{aligned}$$

$$\boxed{S(-\omega) = S(\omega) \;\Rightarrow\; S(\omega) \text{ is an even function of } \omega.}$$

Property 2: $S(\omega)$ is a real function of ω.

Proof. By definition and Euler's formula $e^{-i\omega\tau} = \cos\omega\tau - i\sin\omega\tau$,

$$\begin{aligned} S(\omega) &= \int_{-\infty}^{\infty} R(\tau)\big(\cos\omega\tau - i\sin\omega\tau\big)\,d\tau = \int_{-\infty}^{\infty} R(\tau)\cos\omega\tau\,d\tau - i\int_{-\infty}^{\infty} R(\tau)\sin\omega\tau\,d\tau \\ &= 2\int_{0}^{\infty} R(\tau)\cos\omega\tau\,d\tau - i\cdot 0, \end{aligned}$$

since $R(\tau)\cos\omega\tau$ is even and $R(\tau)\sin\omega\tau$ is odd. Therefore

$$\boxed{S(\omega) = 2\int_{0}^{\infty} R(\tau)\cos\omega\tau\,d\tau \text{ is a real function of } \omega.}$$

Useful working rule. Because $R(\tau)$ and $S(\omega)$ are both even and real,

$$S(\omega) = 2\int_{0}^{\infty} R(\tau)\cos\omega\tau\,d\tau, \qquad R(\tau) = \frac{1}{\pi}\int_{0}^{\infty} S(\omega)\cos\omega\tau\,d\omega.$$

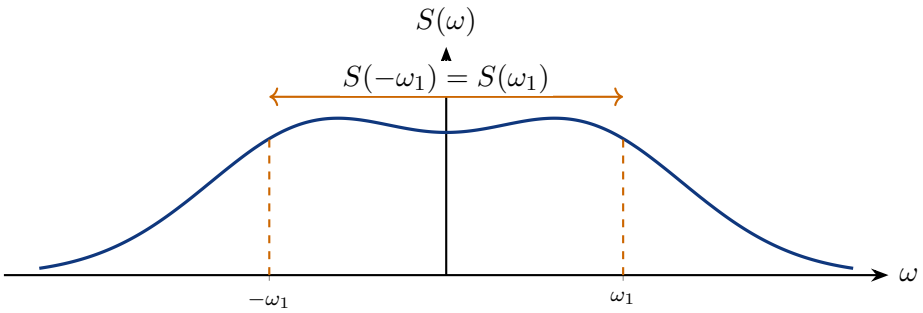


Figure 22: A PSD is real and symmetric about $\omega = 0$; this is why it is enough to work with $\omega \geq 0$ and double.

3.4 Wiener–Khinchine Theorem

Statement

Let $X_T(\omega)$ be the Fourier transform of the truncated random process

$$X_T(t) = \begin{cases} X(t), & -T \leq t \leq T, \\ 0, & \text{elsewhere}, \end{cases}$$

where $X(t)$ is a real WSS process. Then the PSD is given by

$$\boxed{S(\omega) = \lim_{T\to\infty} \frac{1}{2T} E\big[|X_T(\omega)|^2\big].}$$

3.5 Wiener–Khinchine Theorem – Proof

Proof

$X_T(\omega)$ is the Fourier transform of $X_T(t)$:

$$X_T(\omega) = \int_{-\infty}^{\infty} X_T(t) e^{-i\omega t} dt = \int_{-T}^{T} X(t) e^{-i\omega t} dt. \tag{1}$$

Writing the dummy variable as t_1 and as t_2:

$$X_T(\omega) = \int_{-T}^{T} X(t_1) e^{-i\omega t_1} dt_1, \tag{2}$$

$$X_T(\omega) = \int_{-T}^{T} X(t_2) e^{-i\omega t_2} dt_2. \tag{3}$$

Taking the complex conjugate of (3), and using that $X(t)$ is real ($\overline{X(t_2)} = X(t_2)$):

$$\overline{X_T(\omega)} = \int_{-T}^{T} X(t_2) e^{i\omega t_2} dt_2. \tag{4}$$

Multiplying (2) and (4):

$$|X_T(\omega)|^2 = X_T(\omega)\overline{X_T(\omega)} = \int_{-T}^{T}\int_{-T}^{T} X(t_1)X(t_2)\, e^{-i\omega(t_1-t_2)}\, dt_1\, dt_2.$$

Taking expectation:

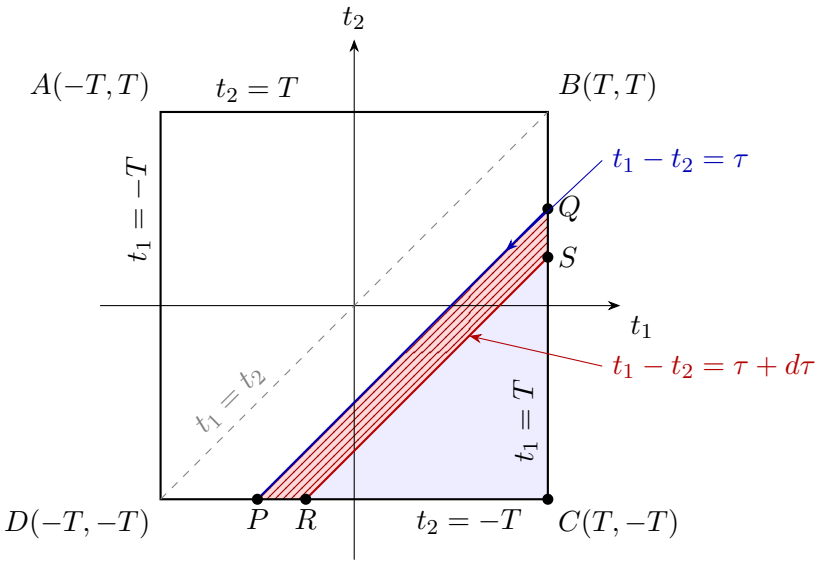
$$E\big[|X_T(\omega)|^2\big] = \int_{-T}^{T}\int_{-T}^{T} E[X(t_1)X(t_2)]\, e^{-i\omega(t_1-t_2)}\, dt_1\, dt_2.$$

Since $X(t)$ is WSS, $E[X(t_1)X(t_2)] = R(\tau)$ with $\tau = t_1 - t_2$:

$$E\big[|X_T(\omega)|^2\big] = \int_{-T}^{T}\int_{-T}^{T} R(\tau)\, e^{-i\omega\tau}\, dt_1\, dt_2. \tag{5}$$

We now convert this double integral into a single integral in τ.

Region of integration



Here t_1 and t_2 each vary from $-T$ to T, so the region is the square $ABCD$, and $\tau = t_1 - t_2$ varies from $-2T$ to $2T$.

For $0 < \tau < 2T$, the lines $t_1 - t_2 = \tau$ and $t_1 - t_2 = \tau + d\tau$ cut off the thin strip $PRSQ$ (hatched) near the corner C.

Strip $PRSQ = \triangle PQC - \triangle RSC$.

Area of the Elementary Strip

$$\begin{aligned} dt_1\, dt_2 &= \text{area of the elementary strip } PRSQ \\ &= \text{area of } \triangle PQC - \text{area of } \triangle RSC \\ &= \tfrac{1}{2}(PC)(QC) - \tfrac{1}{2}(RC)(SC). \end{aligned} \tag{6}$$

P: intersection of $t_1 - t_2 = \tau$ and $t_2 = -T \;\Rightarrow\; t_1 = \tau - T$, i.e. $P(\tau - T,\, -T)$.
Q: intersection of $t_1 - t_2 = \tau$ and $t_1 = T \;\Rightarrow\; t_2 = T - \tau$, i.e. $Q(T,\, T - \tau)$.
With $C(T, -T)$ (right angle at C):

$$PC = T - (\tau - T) = 2T - \tau, \qquad QC = (T - \tau) - (-T) = 2T - \tau.$$

Similarly, R and S lie on $t_1 - t_2 = \tau + d\tau$:

$$R(\tau + d\tau - T,\, -T), \qquad S(T,\, T - \tau - d\tau),$$

$$RC = SC = 2T - (\tau + d\tau) = (2T - \tau) - d\tau.$$

Hence, from (6),

$$dt_1\, dt_2 = \frac{1}{2}(2T-\tau)^2 - \frac{1}{2}\big[(2T-\tau) - d\tau\big]^2.$$

Treat $(2T-\tau)$ as a single term and expand $\big[(2T-\tau)-d\tau\big]^2 = (2T-\tau)^2 - 2(2T-\tau)\,d\tau + (d\tau)^2$.

$$\begin{aligned} dt_1\, dt_2 &= \frac{1}{2}\Big[(2T-\tau)^2 - \Big\{(2T-\tau)^2 - 2(2T-\tau)\,d\tau + (d\tau)^2\Big\}\Big] \\ &= \frac{1}{2}\Big[(2T-\tau)^2 - (2T-\tau)^2 + 2(2T-\tau)\,d\tau - (d\tau)^2\Big] \\ &= \frac{1}{2}\Big[2(2T-\tau)\,d\tau - (d\tau)^2\Big] = (2T-\tau)\,d\tau - \frac{1}{2}(d\tau)^2. \end{aligned}$$

Since $d\tau$ is infinitesimal, $(d\tau)^2$ is of higher order and is neglected:

$$dt_1\, dt_2 = (2T-\tau)\,d\tau, \qquad 0 \le \tau \le 2T.$$

The Case $\tau < 0$

For $-2T \le \tau < 0$, the strip between $t_1 - t_2 = \tau$ and $t_1 - t_2 = \tau + d\tau$ lies near the corner $A(-T, T)$. Its intercepts on $t_1 = -T$ and $t_2 = T$ give legs

$$2T + \tau = 2T - |\tau|,$$

and exactly the same computation gives $dt_1\, dt_2 = (2T+\tau)\,d\tau$.
Combining both cases:

$$\boxed{dt_1\, dt_2 = (2T - |\tau|)\,d\tau, \qquad -2T \le \tau \le 2T.}$$

Substituting into (5):

$$E\big[|X_T(\omega)|^2\big] = \int_{-2T}^{2T} R(\tau)\, e^{-i\omega\tau}\,(2T - |\tau|)\,d\tau.$$

Dividing both sides by $2T$:

$$\begin{aligned} \frac{1}{2T}E\big[|X_T(\omega)|^2\big] &= \int_{-2T}^{2T} R(\tau)\, e^{-i\omega\tau}\left(\frac{2T-|\tau|}{2T}\right)d\tau \\ &= \int_{-2T}^{2T} R(\tau)\, e^{-i\omega\tau}\left(1 - \frac{|\tau|}{2T}\right)d\tau. \end{aligned}$$

Taking the Limit $T \to \infty$

For each fixed τ,

$$\lim_{T\to\infty}\left(1 - \frac{|\tau|}{2T}\right) = 1 - \frac{|\tau|}{\infty} = 1 - 0 = 1,$$

and the limits of integration become $\lim\limits_{T\to\infty}(-2T) = -\infty$, $\lim\limits_{T\to\infty}(2T) = \infty$. Hence (for $R(\tau)$ absolutely integrable)

$$\lim_{T\to\infty}\frac{1}{2T}E\big[|X_T(\omega)|^2\big] = \int_{-\infty}^{\infty} R(\tau)\, e^{-i\omega\tau}\cdot 1\,d\tau = \int_{-\infty}^{\infty} R(\tau)\, e^{-i\omega\tau}\,d\tau.$$

Conclusion
By the definition of the power spectral density,

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau)\, e^{-i\omega\tau}\, d\tau.$$

Therefore

$$\boxed{S(\omega) = \lim_{T\to\infty} \frac{1}{2T}\, E\big[|X_T(\omega)|^2\big]}$$

which completes the proof of the Wiener–Khinchine theorem.

3.6 Problems

Tools used in the solutions

- **Euler's formula:** $e^{\pm i\theta} = \cos\theta \pm i\sin\theta$.
- **Even/odd rule:** on a symmetric interval $(-c, c)$, $\int_{-c}^{c} f = 2\int_{0}^{c} f$ if f is even, and $\int_{-c}^{c} f = 0$ if f is odd.
- **Bernoulli's formula:** $\int u\, v\, dx = u\, v_1 - u'\, v_2 + u''\, v_3 - \cdots$, where $v_1 = \int v$, $v_2 = \int v_1$, $v_3 = \int v_2, \ldots$
- **Cauchy's residue theorem:** $\oint_C f(z)\, dz = 2\pi i \times$ (sum of residues of f at the poles inside C).

Problem 1

Find the PSD if the ACF of the random process $\{X(t)\}$ is

$$R(\tau) = \begin{cases} 1 - \dfrac{|\tau|}{T}, & |\tau| \le T, \\ 0, & \text{elsewhere.} \end{cases}$$

Solution.

Step 1: *Write the definition.*

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau)\, e^{-i\omega\tau}\, d\tau.$$

Step 2: *Split the range according to where $R(\tau)$ is non-zero.* $R(\tau) = 0$ for $\tau < -T$ and for $\tau > T$, so

$$\begin{aligned} S(\omega) &= \int_{-\infty}^{-T} 0 \cdot e^{-i\omega\tau} d\tau + \int_{-T}^{T} \Big(1 - \frac{|\tau|}{T}\Big) e^{-i\omega\tau} d\tau + \int_{T}^{\infty} 0 \cdot e^{-i\omega\tau} d\tau \\ &= \int_{-T}^{T} \Big(1 - \frac{|\tau|}{T}\Big) e^{-i\omega\tau}\, d\tau. \end{aligned}$$

Step 3: *Expand $e^{-i\omega\tau}$ by Euler's formula.*

$$S(\omega) = \int_{-T}^{T} \Big(1 - \frac{|\tau|}{T}\Big) (\cos\omega\tau - i\sin\omega\tau)\, d\tau$$

$$= \int_{-T}^{T} \Big(1 - \frac{|\tau|}{T}\Big) \cos\omega\tau \, d\tau \; - \; i \int_{-T}^{T} \Big(1 - \frac{|\tau|}{T}\Big) \sin\omega\tau \, d\tau.$$

Step 4: *Apply the even/odd rule.* $1 - \frac{|\tau|}{T}$ is even and $\cos\omega\tau$ is even, so their product is even. $1 - \frac{|\tau|}{T}$ is even and $\sin\omega\tau$ is odd, so their product is odd. Hence

$$S(\omega) = 2\int_{0}^{T} \Big(1 - \frac{|\tau|}{T}\Big) \cos\omega\tau \, d\tau - i \cdot 0 = 2\int_{0}^{T} \Big(1 - \frac{|\tau|}{T}\Big) \cos\omega\tau \, d\tau.$$

Step 5: *Remove the modulus.* On $0 \le \tau \le T$, $\tau \ge 0$, so $|\tau| = \tau$:

$$S(\omega) = 2\int_{0}^{T} \Big(1 - \frac{\tau}{T}\Big) \cos\omega\tau \, d\tau.$$

Step 6: *List the terms for Bernoulli's formula.*

$$\begin{aligned} u &= 1 - \frac{\tau}{T}, \quad u' = -\frac{1}{T}, \qquad u'' = 0; \\ v &= \cos\omega\tau, \quad v_1 = \frac{\sin\omega\tau}{\omega}, \quad v_2 = -\frac{\cos\omega\tau}{\omega^2}. \end{aligned}$$

Step 7: *Integrate.*

$$\begin{aligned} \int_{0}^{T} \Big(1 - \frac{\tau}{T}\Big) \cos\omega\tau \, d\tau &= \Big[u\, v_1 - u'\, v_2\Big]_0^T = \Big[\Big(1 - \frac{\tau}{T}\Big)\frac{\sin\omega\tau}{\omega} - \Big(-\frac{1}{T}\Big)\Big(-\frac{\cos\omega\tau}{\omega^2}\Big)\Big]_0^T \\ &= \Big[\Big(1 - \frac{\tau}{T}\Big)\frac{\sin\omega\tau}{\omega} - \frac{\cos\omega\tau}{\omega^2 T}\Big]_0^T. \end{aligned}$$

Step 8: *Substitute the limits.*

$$\begin{aligned} \text{At } \tau = T: \quad & (1-1)\frac{\sin\omega T}{\omega} - \frac{\cos\omega T}{\omega^2 T} = -\frac{\cos\omega T}{\omega^2 T}. \\ \text{At } \tau = 0: \quad & (1-0)\frac{\sin 0}{\omega} - \frac{\cos 0}{\omega^2 T} = -\frac{1}{\omega^2 T}. \\ \text{Upper} - \text{lower}: \quad & -\frac{\cos\omega T}{\omega^2 T} + \frac{1}{\omega^2 T} = \frac{1 - \cos\omega T}{\omega^2 T}. \end{aligned}$$

Step 9: *Multiply by 2 and simplify.*

$$S(\omega) = 2 \cdot \frac{1 - \cos\omega T}{\omega^2 T} = \frac{2(1 - \cos\omega T)}{\omega^2 T}.$$

Using $1 - \cos\omega T = 2\sin^2\dfrac{\omega T}{2}$:

$$S(\omega) = \frac{4\sin^2(\omega T/2)}{\omega^2 T} = T \cdot \frac{\sin^2(\omega T/2)}{(\omega T/2)^2}.$$

$$\boxed{S(\omega) = \frac{2}{\omega^2 T}\big(1 - \cos\omega T\big) = T\left[\frac{\sin(\omega T/2)}{\omega T/2}\right]^2}$$

Check: $S(\omega) \ge 0$ for every ω, as a PSD must be, and $S(\omega) \to T$ as $\omega \to 0$.

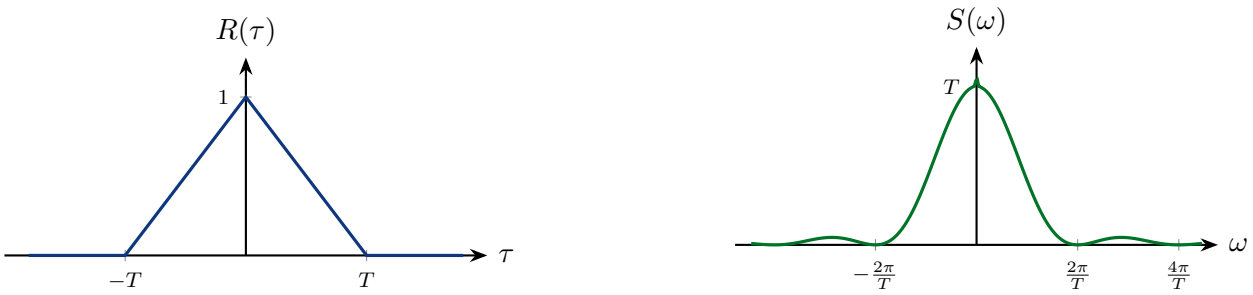


Figure 23: Triangular ACF (left) and its PSD (right). The PSD vanishes at $\omega = 2n\pi/T$, $n \neq 0$.

Problem 2

Find the ACF $R(\tau)$ of a stationary random process $\{X(t)\}$ whose PSD is

$$S(\omega) = \begin{cases} 1+\omega^2, & |\omega| \le 1 \quad (-1 \le \omega \le 1), \\ 0, & \text{elsewhere.} \end{cases}$$

Solution.

Step 1: *Write the inverse relation.*

$$R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\, e^{i\omega\tau}\, d\omega.$$

Step 2: *Keep only the range where $S(\omega) \neq 0$.*

$$R(\tau) = \frac{1}{2\pi}\left(\int_{-\infty}^{-1} 0\, d\omega + \int_{-1}^{1}(1+\omega^2)e^{i\omega\tau}d\omega + \int_{1}^{\infty} 0\, d\omega\right) = \frac{1}{2\pi}\int_{-1}^{1}(1+\omega^2)\, e^{i\omega\tau}\, d\omega.$$

Step 3: *Use Euler's formula.*

$$R(\tau) = \frac{1}{2\pi}\int_{-1}^{1}(1+\omega^2)\cos\omega\tau\, d\omega + \frac{i}{2\pi}\int_{-1}^{1}(1+\omega^2)\sin\omega\tau\, d\omega.$$

Step 4: *Even/odd rule (now in the variable ω).* $(1+\omega^2)\cos\omega\tau$ is even in ω, and $(1+\omega^2)\sin\omega\tau$ is odd in ω. So

$$R(\tau) = \frac{1}{2\pi}\cdot 2\int_{0}^{1}(1+\omega^2)\cos\omega\tau\, d\omega + 0 = \frac{1}{\pi}\int_{0}^{1}(1+\omega^2)\cos\omega\tau\, d\omega.$$

Step 5: *List the terms for Bernoulli's formula (integrating with respect to ω; τ is a constant).*

$$u = 1+\omega^2, \quad u' = 2\omega, \qquad u'' = 2, \qquad u''' = 0;$$
$$v = \cos\omega\tau, \quad v_1 = \frac{\sin\omega\tau}{\tau}, \quad v_2 = -\frac{\cos\omega\tau}{\tau^2}, \quad v_3 = -\frac{\sin\omega\tau}{\tau^3}.$$

Step 6: *Integrate.*

$$\begin{aligned}\int_{0}^{1}(1+\omega^2)\cos\omega\tau\, d\omega &= \Big[u\,v_1 - u'\,v_2 + u''\,v_3\Big]_0^1 \\ &= \left[(1+\omega^2)\frac{\sin\omega\tau}{\tau} - (2\omega)\Big(-\frac{\cos\omega\tau}{\tau^2}\Big) + 2\Big(-\frac{\sin\omega\tau}{\tau^3}\Big)\right]_0^1\end{aligned}$$

$$= \left[(1+\omega^2)\frac{\sin\omega\tau}{\tau} + \frac{2\omega\cos\omega\tau}{\tau^2} - \frac{2\sin\omega\tau}{\tau^3}\right]_0^1.$$

Step 7: *Substitute the limits.*

$$\text{At } \omega = 1: \quad (1+1)\frac{\sin\tau}{\tau} + \frac{2\cos\tau}{\tau^2} - \frac{2\sin\tau}{\tau^3} = \frac{2\sin\tau}{\tau} + \frac{2\cos\tau}{\tau^2} - \frac{2\sin\tau}{\tau^3}.$$
$$\text{At } \omega = 0: \quad (1)\frac{\sin 0}{\tau} + \frac{0}{\tau^2} - \frac{2\sin 0}{\tau^3} = 0.$$

Step 8: *Multiply by* $\frac{1}{\pi}$ *and take the common denominator* τ^3.

$$R(\tau) = \frac{1}{\pi}\left[\frac{2\sin\tau}{\tau} + \frac{2\cos\tau}{\tau^2} - \frac{2\sin\tau}{\tau^3}\right] = \frac{1}{\pi}\cdot\frac{2\tau^2\sin\tau + 2\tau\cos\tau - 2\sin\tau}{\tau^3}$$
$$= \frac{2}{\pi\tau^3}\Big(\tau\cos\tau + \tau^2\sin\tau - \sin\tau\Big).$$

$$\boxed{R(\tau) = \frac{2}{\pi\tau^3}\Big(\tau\cos\tau + (\tau^2-1)\sin\tau\Big)}$$

Check: as $\tau \to 0$, $R(\tau) \to \frac{1}{2\pi}\int_{-1}^{1}(1+\omega^2)\,d\omega = \frac{1}{2\pi}\cdot\frac{8}{3} = \frac{4}{3\pi}$, the average power.

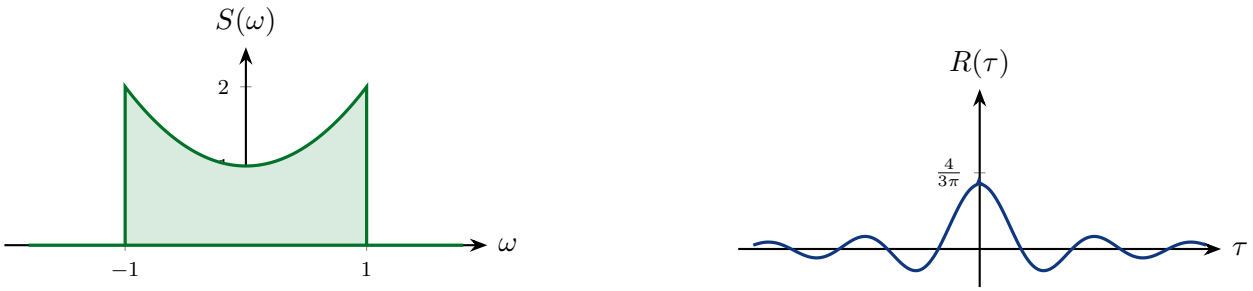


Figure 24: Problem 2: the band-limited PSD (left) and the resulting oscillating ACF (right).

Problem 3

Find the ACF $R(\tau)$ when the PSD of a random process $\{X(t)\}$ is given by

$$S(\omega) = \begin{cases} 1, & |\omega| \le \omega_0, \\ 0, & \text{elsewhere.} \end{cases}$$

Hence prove that $X(t)$ and $X\big(t + \frac{\pi}{\omega_0}\big)$ are uncorrelated random variables, where $\{X(t)\}$ is a zero-mean WSS process.

Solution.

Part A: the ACF.

Step 1: *Inverse relation, restricted to* $|\omega| \le \omega_0$.

$$R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)e^{i\omega\tau}d\omega = \frac{1}{2\pi}\int_{-\omega_0}^{\omega_0} 1\cdot e^{i\omega\tau}\,d\omega.$$

Step 2: *Integrate the exponential directly.*

$$\int_{-\omega_0}^{\omega_0} e^{i\omega\tau}d\omega = \left[\frac{e^{i\omega\tau}}{i\tau}\right]_{-\omega_0}^{\omega_0} = \frac{e^{i\omega_0\tau} - e^{-i\omega_0\tau}}{i\tau}.$$

Step 3: *Use* $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$.

$$\frac{e^{i\omega_0\tau} - e^{-i\omega_0\tau}}{i\tau} = \frac{2i\sin\omega_0\tau}{i\tau} = \frac{2\sin\omega_0\tau}{\tau}.$$

Step 4: *Multiply by* $\frac{1}{2\pi}$.

$$R(\tau) = \frac{1}{2\pi}\cdot\frac{2\sin\omega_0\tau}{\tau} = \frac{1}{\pi}\frac{\sin\omega_0\tau}{\tau}.$$

$$\boxed{R(\tau) = \frac{1}{\pi}\frac{\sin\omega_0\tau}{\tau}}$$

(The cosine form gives the same: $\frac{1}{\pi}\int_0^{\omega_0}\cos\omega\tau\,d\omega = \frac{1}{\pi}\left[\frac{\sin\omega\tau}{\tau}\right]_0^{\omega_0}$.)

Part B: uncorrelated samples.

Step 5: *State what must be shown.* $X(t)$ and $X\left(t+\frac{\pi}{\omega_0}\right)$ are uncorrelated if and only if

$$\mathrm{Cov}\left(X(t), X(t+\tfrac{\pi}{\omega_0})\right) = 0.$$

Step 6: *Write the covariance.* Since $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$,

$$\mathrm{Cov}\left(X(t), X\left(t+\tfrac{\pi}{\omega_0}\right)\right) = E\left(X(t)\,X\left(t+\tfrac{\pi}{\omega_0}\right)\right) - E\big(X(t)\big)\,E\left(X\left(t+\tfrac{\pi}{\omega_0}\right)\right).$$

Step 7: *Use zero mean.* $E\big(X(t)\big) = 0$ for all t, so the second term vanishes:

$$\mathrm{Cov} = E\left(X(t)\,X\left(t+\tfrac{\pi}{\omega_0}\right)\right).$$

Step 8: *Recognise the ACF.* Writing $t+\frac{\pi}{\omega_0} = t-\left(-\frac{\pi}{\omega_0}\right)$ and using $R(\tau) = E[X(t)X(t-\tau)]$ (equivalently, R is even):

$$\mathrm{Cov} = R\Big(-\frac{\pi}{\omega_0}\Big) = R\Big(\frac{\pi}{\omega_0}\Big).$$

Step 9: *Substitute* $\tau = \pi/\omega_0$ *in the ACF.*

$$R\Big(\frac{\pi}{\omega_0}\Big) = \frac{1}{\pi}\cdot\frac{\sin\left(\omega_0\cdot\frac{\pi}{\omega_0}\right)}{\pi/\omega_0} = \frac{1}{\pi}\cdot\frac{\sin\pi}{\pi/\omega_0} = \frac{1}{\pi}\cdot\frac{0}{\pi/\omega_0} = 0.$$

$$\boxed{\mathrm{Cov}\left(X(t), X(t+\pi/\omega_0)\right) = 0 \;\Rightarrow \text{they are uncorrelated random variables.}}$$

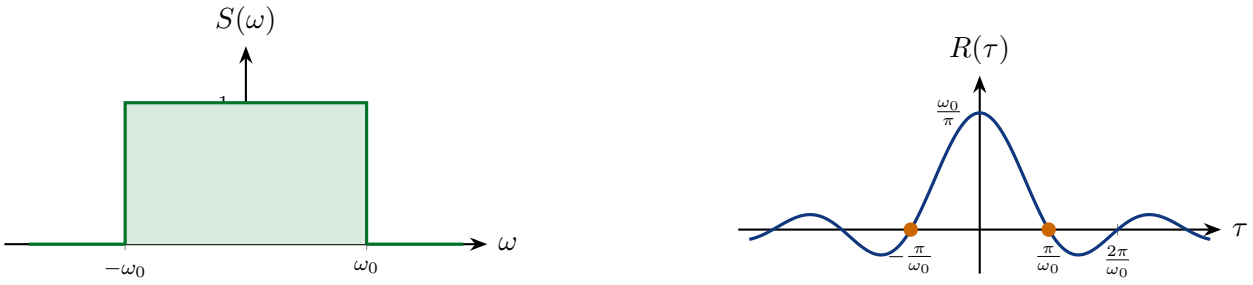


Figure 25: Band-limited ("ideal low-pass") PSD and its ACF. The ACF first crosses zero at $\tau = \pm\pi/\omega_0$ (amber dots): samples that far apart are uncorrelated.

Problem 4

Find the average power of the random process $\{X(t)\}$ if its power spectral density is

$$S(\omega) = \frac{10\omega^2 + 35}{(\omega^2 + 4)(\omega^2 + 9)}$$

(i) using $S(\omega)$ directly, (ii) using the autocorrelation function $R(\tau)$.

Solution. The average power of $\{X(t)\}$ is

$$R(0) = E\big(X^2(t)\big) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\, d\omega.$$

(i) Using $S(\omega)$ directly.

Step 1: *Write the integral.*

$$R(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{10\omega^2 + 35}{(\omega^2 + 4)(\omega^2 + 9)}\, d\omega.$$

Step 2: *Convert to a contour integral.* Replace ω by the complex variable z and integrate over C, the real axis from $-R$ to R together with the large semicircle Γ in the upper half-plane. As $R \to \infty$, the integral over Γ tends to 0 (the denominator has degree 4, two more than the numerator), so

$$R(0) = \frac{1}{2\pi}\oint_C f(z)\, dz, \qquad f(z) = \frac{10z^2 + 35}{(z^2 + 4)(z^2 + 9)}.$$

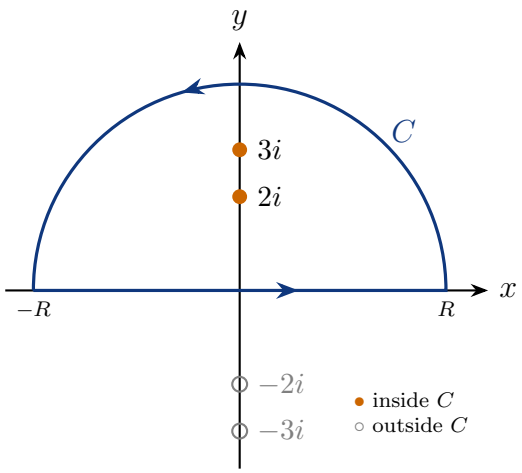


Figure 26: Contour C: the real axis from $-R$ to R closed by a large semicircle in the upper half-plane. Only the poles $2i$ and $3i$ lie inside C.

Step 3: *Apply Cauchy's residue theorem.*

$$R(0) = \frac{1}{2\pi} \cdot 2\pi i \,\big[\text{sum of residues inside } C\big],$$

$$\Longrightarrow \quad R(0) = i\,\big[\text{sum of residues inside } C\big] \qquad \cdots (1)$$

Step 4: *Find the poles.* Equate the denominator to zero:

$$(z^2 + 4)(z^2 + 9) = 0 \;\Rightarrow\; z^2 = -4 \text{ or } z^2 = -9 \;\Rightarrow\; z = \pm 2i,\ \pm 3i.$$

Each is a simple pole (order 1). Only $z = 2i$ and $z = 3i$ (positive imaginary part) lie inside C.

Step 5: *Residue at $z = 2i$.* Factorise $z^2 + 4 = (z + 2i)(z - 2i)$ and cancel $(z - 2i)$:

$$\big(\operatorname{Res} f\big)_{z=2i} = \lim_{z\to 2i}(z - 2i)\,\frac{10z^2 + 35}{(z + 2i)(z - 2i)(z^2 + 9)} = \lim_{z\to 2i}\frac{10z^2 + 35}{(z + 2i)(z^2 + 9)}.$$

At $z = 2i$: $z^2 = -4$, so $10z^2 + 35 = -40 + 35 = -5$; $z + 2i = 4i$; $z^2 + 9 = -4 + 9 = 5$. Therefore

$$\big(\operatorname{Res} f\big)_{z=2i} = \frac{-5}{(4i)(5)} = \frac{-5}{20i} = -\frac{1}{4i}.$$

Step 6: *Residue at $z = 3i$.* Factorise $z^2 + 9 = (z + 3i)(z - 3i)$ and cancel $(z - 3i)$:

$$\big(\operatorname{Res} f\big)_{z=3i} = \lim_{z\to 3i}\frac{10z^2 + 35}{(z^2 + 4)(z + 3i)}.$$

At $z = 3i$: $z^2 = -9$, so $10z^2 + 35 = -90 + 35 = -55$; $z^2 + 4 = -9 + 4 = -5$; $z + 3i = 6i$. Therefore

$$\big(\operatorname{Res} f\big)_{z=3i} = \frac{-55}{(-5)(6i)} = \frac{-55}{-30i} = \frac{11}{6i}.$$

Step 7: *Substitute in (1).*

$$R(0) = i\left[-\frac{1}{4i} + \frac{11}{6i}\right] = -\frac{i}{4i} + \frac{11i}{6i} = -\frac{1}{4} + \frac{11}{6} = \frac{-3 + 22}{12} = \frac{19}{12}.$$

$$R(0) = \frac{19}{12} = \text{average power of the random process } \{X(t)\}.$$

(ii) Using the ACF $R(\tau)$.

Step 1: *Write the inverse relation.*

$$R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{10\omega^2 + 35}{(\omega^2 + 4)(\omega^2 + 9)}\,e^{i\omega\tau}\,d\omega.$$

Step 2: *Contour integral (take $\tau > 0$).* With the same contour C (for $\tau > 0$, $|e^{iz\tau}| = e^{-\tau \operatorname{Im} z} \le 1$ in the upper half-plane, so the semicircle contributes nothing in the limit),

$$R(\tau) = \frac{1}{2\pi}\oint_C f(z)\,dz, \qquad f(z) = \frac{(10z^2 + 35)\,e^{iz\tau}}{(z^2 + 4)(z^2 + 9)},$$

$$R(\tau) = i\,\big[\text{sum of residues of } f(z) \text{ inside } C\big] \qquad \cdots(2)$$

Step 3: *Poles inside C.* The factor $e^{iz\tau}$ never vanishes and has no poles, so the poles inside C are again $z = 2i$, $3i$, both simple.

Step 4: *Residues.* The work is the same as in (i), with the extra factor $e^{iz\tau}$ evaluated at the pole:

$$\big(\operatorname{Res} f\big)_{z=2i} = -\frac{1}{4i}\,e^{i(2i)\tau} = -\frac{1}{4i}\,e^{i^2\cdot 2\tau} = -\frac{1}{4i}\,e^{-2\tau},$$
$$\big(\operatorname{Res} f\big)_{z=3i} = \frac{11}{6i}\,e^{i(3i)\tau} = \frac{11}{6i}\,e^{i^2\cdot 3\tau} = \frac{11}{6i}\,e^{-3\tau}.$$

Step 5: *Substitute in (2).*

$$R(\tau) = i\left(-\frac{1}{4i}e^{-2\tau} + \frac{11}{6i}e^{-3\tau}\right) = \frac{11}{6}e^{-3\tau} - \frac{1}{4}e^{-2\tau}, \qquad \tau \ge 0.$$

Step 6: *Extend to all τ.* $R(\tau)$ is even, so replace τ by $|\tau|$:

$$R(\tau) = \frac{11}{6}e^{-3|\tau|} - \frac{1}{4}e^{-2|\tau|}.$$

Step 7: *Put $\tau = 0$.*

$$R(0) = \frac{11}{6}e^{0} - \frac{1}{4}e^{0} = \frac{11}{6} - \frac{1}{4} = \frac{22-3}{12} = \frac{19}{12}.$$

$$\boxed{R(\tau) = \frac{11}{6}e^{-3|\tau|} - \frac{1}{4}e^{-2|\tau|}, \qquad \text{average power } R(0) = \frac{19}{12}.}$$

Cross-check by partial fractions: with $u = \omega^2$, $\dfrac{10u+35}{(u+4)(u+9)} = \dfrac{A}{u+4} + \dfrac{B}{u+9}$ gives $A = -1$, $B = 11$; using the pair $e^{-a|\tau|} \leftrightarrow \dfrac{2a}{\omega^2+a^2}$ gives the same $R(\tau)$.

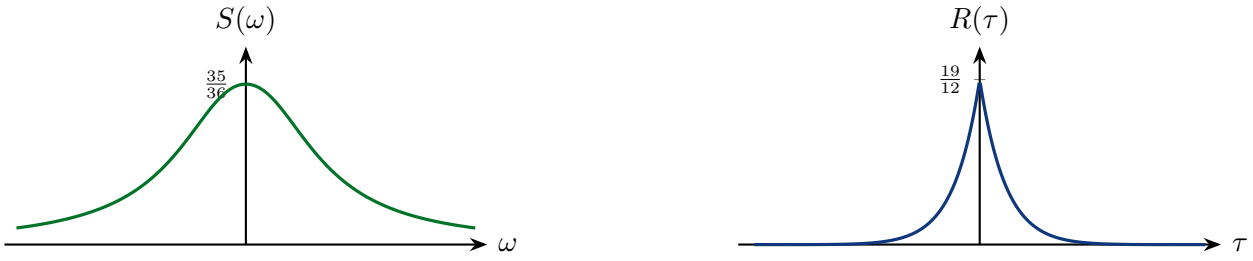


Figure 27: Problem 4: rational PSD (left) and its ACF (right); the peak of the ACF is the average power 19/12.

Problem 5

The power spectrum of a WSS process $\{X(t)\}$ is given by $S(\omega) = \dfrac{1}{(1+\omega^2)^2}$. Find its ACF and average power.

Solution.

Step 1: *Inverse relation as a contour integral ($\tau > 0$).*

$$R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{e^{i\omega\tau}}{(1+\omega^2)^2}\,d\omega = \frac{1}{2\pi}\oint_C f(z)\,dz, \qquad f(z) = \frac{e^{iz\tau}}{(z^2+1)^2},$$

where C is the large upper semicircle. By the residue theorem,

$$R(\tau) = i\,[\text{sum of residues inside } C].$$

Step 2: *Find the poles and their order.* $z^2+1 = (z-i)(z+i)$, so

$$f(z) = \frac{e^{iz\tau}}{(z-i)^2(z+i)^2}.$$

The poles are $z = i$ and $z = -i$, each of order 2. Only $z = i$ lies inside C.

Step 3: *Residue formula for a pole of order n.*

$$\big(\operatorname{Res} f\big)_{z=a} = \frac{1}{(n-1)!}\lim_{z\to a}\frac{d^{n-1}}{dz^{n-1}}\Big[(z-a)^n f(z)\Big].$$

Here $a = i$ and $n = 2$, so

$$\big(\operatorname{Res} f\big)_{z=i} = \frac{1}{1!}\lim_{z\to i}\frac{d}{dz}\Big[(z-i)^2\frac{e^{iz\tau}}{(z-i)^2(z+i)^2}\Big] = \lim_{z\to i}\frac{d}{dz}\left[\frac{e^{iz\tau}}{(z+i)^2}\right].$$

Step 4: *Differentiate (product rule on $e^{iz\tau}(z+i)^{-2}$).*

$$\frac{d}{dz}\Big[e^{iz\tau}(z+i)^{-2}\Big] = i\tau\, e^{iz\tau}(z+i)^{-2} + e^{iz\tau}\cdot(-2)(z+i)^{-3} = e^{iz\tau}\left[\frac{i\tau}{(z+i)^2} - \frac{2}{(z+i)^3}\right].$$

Step 5: *Put $z = i$.* We need $e^{i\cdot i\tau} = e^{-\tau}$, $z+i = 2i$, $(2i)^2 = -4$, $(2i)^3 = 8i^3 = -8i$:

$$\big(\operatorname{Res} f\big)_{z=i} = e^{-\tau}\left[\frac{i\tau}{-4} - \frac{2}{-8i}\right] = e^{-\tau}\left[-\frac{i\tau}{4} + \frac{1}{4i}\right].$$

Step 6: *Simplify $\frac{1}{4i}$.* Since $\frac{1}{i} = \frac{i}{i^2} = -i$, we get $\frac{1}{4i} = -\frac{i}{4}$. Hence

$$\big(\operatorname{Res} f\big)_{z=i} = e^{-\tau}\left[-\frac{i\tau}{4} - \frac{i}{4}\right] = -\frac{i}{4}(1+\tau)\,e^{-\tau}.$$

Step 7: *Multiply by i.*

$$R(\tau) = i\cdot\Big(-\frac{i}{4}(1+\tau)e^{-\tau}\Big) = -\frac{i^2}{4}(1+\tau)e^{-\tau} = \frac{1}{4}(1+\tau)e^{-\tau}, \qquad \tau \ge 0.$$

Step 8: *Extend to all τ and find the average power.* Since R is even,

$$R(\tau) = \frac{1}{4}\big(1+|\tau|\big)e^{-|\tau|}, \qquad R(0) = \frac{1}{4}(1+0)e^{0} = \frac{1}{4}.$$

$$\boxed{R(\tau) = \frac{1}{4}\big(1+|\tau|\big)e^{-|\tau|}, \qquad \text{Average power } = R(0) = \frac{1}{4}.}$$

Check: $\dfrac{1}{2\pi}\displaystyle\int_{-\infty}^{\infty}\frac{d\omega}{(1+\omega^2)^2} = \frac{1}{2\pi}\cdot\frac{\pi}{2} = \frac{1}{4}.$

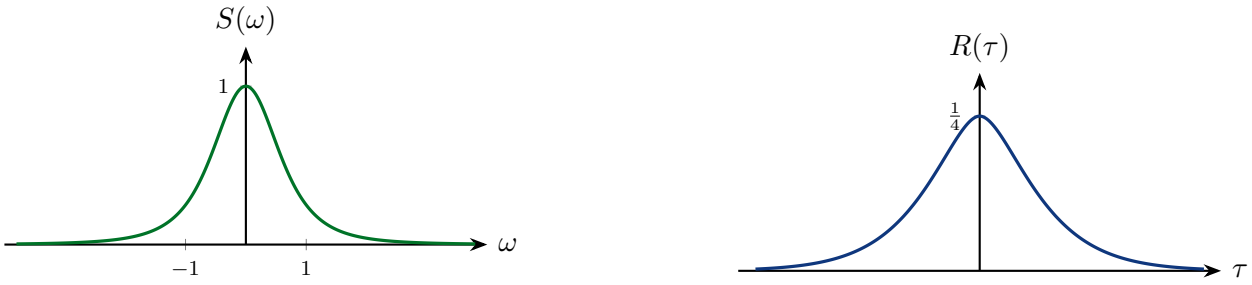


Figure 28: Problem 5: $S(\omega) = 1/(1+\omega^2)^2$ and $R(\tau) = \frac{1}{4}(1+|\tau|)e^{-|\tau|}$.

3.7 PSD of the Random Telegraph Signal

The random telegraph signal $\{X(t)\}$ takes only two values, $+a$ and $-a$, and switches between them at random instants. A narrow (fast-decaying) ACF means the signal "forgets" its past quickly.

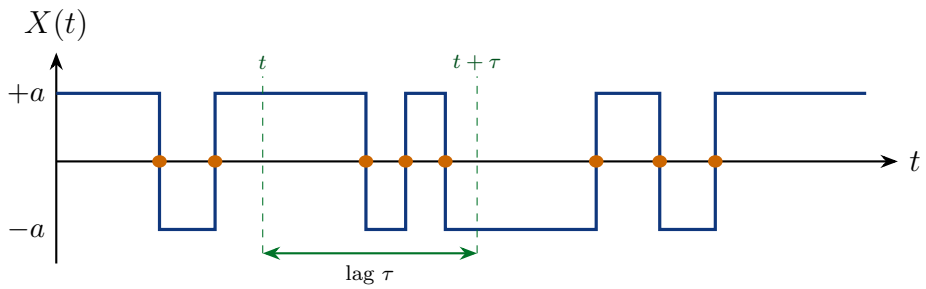


Figure 29: A sample path of the random telegraph signal with levels $\pm a$. Amber dots mark the random switching instants.

Example 1 **(BDU — Apr. 97)**

The autocorrelation function of the random telegraph signal process is given by $R(\tau) = a^2 e^{-2\gamma|\tau|}$. Determine the power density spectrum of the random telegraph signal.

Solution. Here a and $\gamma > 0$ are constants.

Method 1: splitting the modulus

Step 1: *Write the definition.*

$$S(\omega) = \int_{-\infty}^{\infty} R(\tau)\, e^{-i\omega\tau}\, d\tau = \int_{-\infty}^{\infty} a^2 e^{-2\gamma|\tau|}\, e^{-i\omega\tau}\, d\tau.$$

Step 2: *Remove the modulus.* $|\tau| = -\tau$ for $\tau < 0$ and $|\tau| = \tau$ for $\tau \geq 0$. So

$$e^{-2\gamma|\tau|} = \begin{cases} e^{2\gamma\tau}, & \tau < 0, \\ e^{-2\gamma\tau}, & \tau \geq 0. \end{cases}$$

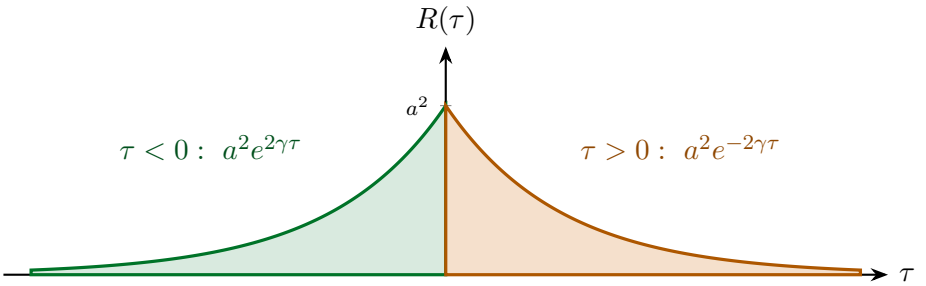


Figure 30: The ACF $a^2 e^{-2\gamma|\tau|}$ is made of two exponentials: $a^2 e^{2\gamma\tau}$ on the left of $\tau = 0$ and $a^2 e^{-2\gamma\tau}$ on the right.

Step 3: *Split the integral at* $\tau = 0$.

$$S(\omega) = a^2 \int_{-\infty}^{0} e^{2\gamma\tau} e^{-i\omega\tau}\, d\tau + a^2 \int_{0}^{\infty} e^{-2\gamma\tau} e^{-i\omega\tau}\, d\tau.$$

Step 4: *Combine the exponents.*

$$S(\omega) = a^2 \int_{-\infty}^{0} e^{(2\gamma - i\omega)\tau}\, d\tau + a^2 \int_{0}^{\infty} e^{-(2\gamma + i\omega)\tau}\, d\tau.$$

Step 5: *Integrate each term* (using $\int e^{k\tau} d\tau = \frac{e^{k\tau}}{k}$):

$$S(\omega) = a^2 \left[\frac{e^{(2\gamma - i\omega)\tau}}{2\gamma - i\omega} \right]_{-\infty}^{0} + a^2 \left[\frac{e^{-(2\gamma + i\omega)\tau}}{-(2\gamma + i\omega)} \right]_{0}^{\infty}.$$

Step 6: *Evaluate the limits.* Since $|e^{\mp i\omega\tau}| = 1$ and $\gamma > 0$: $e^{2\gamma\tau} \to 0$ as $\tau \to -\infty$, and $e^{-2\gamma\tau} \to 0$ as $\tau \to \infty$. Hence

$$\left[\frac{e^{(2\gamma - i\omega)\tau}}{2\gamma - i\omega} \right]_{-\infty}^{0} = \frac{e^0}{2\gamma - i\omega} - 0 = \frac{1}{2\gamma - i\omega},$$

$$\left[\frac{e^{-(2\gamma + i\omega)\tau}}{-(2\gamma + i\omega)} \right]_{0}^{\infty} = 0 - \frac{e^0}{-(2\gamma + i\omega)} = \frac{1}{2\gamma + i\omega}.$$

Step 7: *Add the two fractions.*

$$S(\omega) = a^2 \left[\frac{1}{2\gamma - i\omega} + \frac{1}{2\gamma + i\omega} \right] = a^2 \left[\frac{(2\gamma + i\omega) + (2\gamma - i\omega)}{(2\gamma - i\omega)(2\gamma + i\omega)} \right].$$

Step 8: *Simplify.* Numerator: $(2\gamma + i\omega) + (2\gamma - i\omega) = 4\gamma$. Denominator: $(2\gamma)^2 - (i\omega)^2 = 4\gamma^2 - i^2\omega^2 = 4\gamma^2 + \omega^2$. Therefore

$$S(\omega) = a^2 \cdot \frac{4\gamma}{4\gamma^2 + \omega^2}.$$

$$\boxed{S(\omega) = \frac{4a^2\gamma}{\omega^2 + 4\gamma^2}}$$

Method 2: cosine form (Property 2)

Step 1: *Use the cosine form, since $R(\tau)$ is even.*

$$S(\omega) = 2 \int_{0}^{\infty} R(\tau) \cos\omega\tau\, d\tau = 2a^2 \int_{0}^{\infty} e^{-2\gamma\tau} \cos\omega\tau\, d\tau.$$

Step 2: *Use the standard integral* $\int e^{\alpha\tau} \cos\beta\tau\, d\tau = \frac{e^{\alpha\tau}}{\alpha^2 + \beta^2}\big(\alpha\cos\beta\tau + \beta\sin\beta\tau\big)$, with $\alpha = -2\gamma$, $\beta = \omega$:

$$\int_{0}^{\infty} e^{-2\gamma\tau} \cos\omega\tau\, d\tau = \left[\frac{e^{-2\gamma\tau}}{4\gamma^2 + \omega^2} \big(-2\gamma\cos\omega\tau + \omega\sin\omega\tau \big) \right]_{0}^{\infty}.$$

Step 3: *Evaluate the limits.* At $\tau \to \infty$ the factor $e^{-2\gamma\tau} \to 0$, so the value is 0. At $\tau = 0$: $\frac{1}{4\gamma^2 + \omega^2}(-2\gamma \cdot 1 + 0) = \frac{-2\gamma}{4\gamma^2 + \omega^2}$. Hence

$$\int_{0}^{\infty} e^{-2\gamma\tau} \cos\omega\tau\, d\tau = 0 - \frac{-2\gamma}{4\gamma^2 + \omega^2} = \frac{2\gamma}{4\gamma^2 + \omega^2}.$$

Step 4: *Multiply by $2a^2$.*

$$S(\omega) = 2a^2 \cdot \frac{2\gamma}{4\gamma^2 + \omega^2} = \frac{4a^2\gamma}{\omega^2 + 4\gamma^2},$$

the same result as Method 1.

Verification: average power

From the ACF: $R(0) = a^2 e^0 = a^2$. From the PSD:

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{4a^2\gamma}{\omega^2 + (2\gamma)^2}\, d\omega = \frac{4a^2\gamma}{2\pi} \cdot \frac{1}{2\gamma}\Big[\tan^{-1}\frac{\omega}{2\gamma}\Big]_{-\infty}^{\infty} = \frac{a^2}{\pi}\Big[\frac{\pi}{2} - \Big(-\frac{\pi}{2}\Big)\Big] = \frac{a^2}{\pi}\cdot\pi = a^2.$$

Both agree: the average power of the telegraph signal is a^2 (indeed $X^2(t) = a^2$ always).

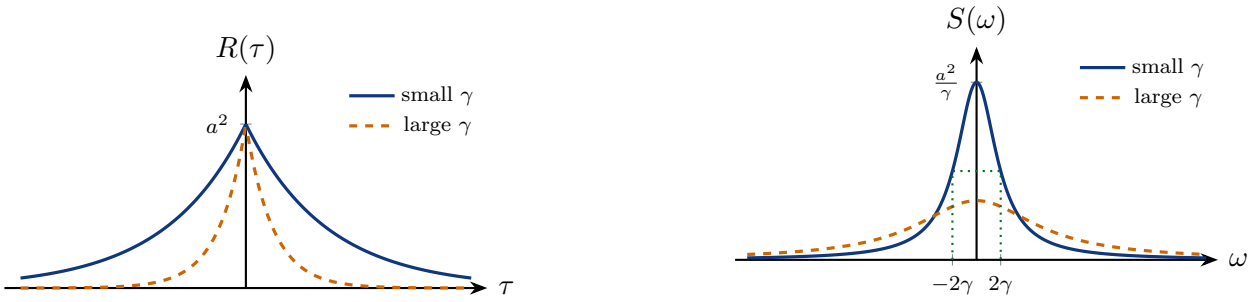


Figure 31: Telegraph signal: a larger γ (faster switching) makes the ACF decay faster and spreads the PSD over a wider band. The dotted line marks the half-power point $\omega = 2\gamma$ for the small-γ curve.

Interpretation

- Peak value $S(0) = \dfrac{4a^2\gamma}{4\gamma^2} = \dfrac{a^2}{\gamma}$.
- Half-power frequency: $S(\omega) = \frac{1}{2}S(0) \iff \omega^2 + 4\gamma^2 = 8\gamma^2 \iff \omega = 2\gamma$. The bandwidth grows in proportion to γ.
- A narrow ACF (short memory) goes with a wide PSD, and a wide ACF with a narrow PSD; the total power a^2 is the same in both.

3.8 Summary

ACF $R(\tau)$	**PSD** $S(\omega)$
$1 - \dfrac{\lvert\tau\rvert}{T},\ \lvert\tau\rvert \le T$	$\dfrac{2}{\omega^2 T}(1 - \cos\omega T)$
$\dfrac{2}{\pi\tau^3}\big(\tau\cos\tau + (\tau^2 - 1)\sin\tau\big)$	$1 + \omega^2,\ \lvert\omega\rvert \le 1$
$\dfrac{\sin\omega_0\tau}{\pi\tau}$	$1,\ \lvert\omega\rvert \le \omega_0$
$\dfrac{11}{6}e^{-3\lvert\tau\rvert} - \dfrac{1}{4}e^{-2\lvert\tau\rvert}$	$\dfrac{10\omega^2 + 35}{(\omega^2+4)(\omega^2+9)}$
$\dfrac{1}{4}(1 + \lvert\tau\rvert)e^{-\lvert\tau\rvert}$	$\dfrac{1}{(1+\omega^2)^2}$
$a^2 e^{-2\gamma\lvert\tau\rvert}$ (telegraph signal)	$\dfrac{4a^2\gamma}{\omega^2 + 4\gamma^2}$

4 Assignment – 4

Questions and Hints

4.1 Questions

Question 1

Verify whether the cosine wave process

$$X(t) = Y \cos \omega t,$$

where Y is a uniformly distributed random variable in $(0, 1)$, is an SSS process.

Question 2

If $X(t) = Y \cos \omega t + Z \sin \omega t$, where Y and Z are independent random variables with mean zero and the same variance, prove that $X(t)$ is an SSS process of order 2, i.e. a WSS process.

Question 3

Given the random variable Y with characteristic function

$$\phi(\omega) = E\left(e^{i\omega Y}\right) = E(\cos \omega Y + i \sin \omega Y)$$

and a random process defined by $X(t) = \cos(\lambda t + Y)$, show that $\{X(t)\}$ is stationary in the wide sense if $\phi(1) = \phi(2) = 0$.

Question 4

In a fair coin experiment, we define the process $\{X(t)\}$ as follows:

$$X(t) = \begin{cases} \sin \pi t, & \text{if head shows,} \\ 2t, & \text{if tail shows.} \end{cases}$$

Find $E[X(t)]$.

Question 5

A radio station broadcasts its signals in the form of a cosine wave. Receivers receive its signals from different places and its collection is given by

$$X(t) = 10 \cos(100t + \theta).$$

Based on the distance of the receivers, the phase difference θ is treated as a random variable which follows the uniform distribution in $(-\pi, \pi)$. Prove that $X(t)$ is a WSS process.

Question 6

Prove that the spectral density and the autocorrelation function of a real WSS process form a Fourier cosine transform pair.

Question 7

Find the power spectral density function of a random process $\{X(t)\}$ whose ACF is given by

$$R(\tau) = e^{-\alpha\tau^2}, \qquad -\infty < \tau < \infty.$$

Hence find $S(\omega)$ when $R(\tau) = e^{-\alpha\tau^2}\cos\omega_0\tau$.

Question 8

Find the ACF $R(\tau)$ for a WSS random process $\{X(t)\}$ whose PSD function $S(\omega)$ is given by

$$S(\omega) = \frac{1}{(1+\omega^2)^2}.$$

Also find the average power of the random process $\{X(t)\}$. $\big(R(0) = E[X^2(t)]\big)$

Question 9

If $X(t) = R\cos(\omega t + \phi)$, where R and ϕ are independent random variables and ϕ is uniformly distributed in $(-\pi, \pi)$, find $R(t_1, t_2)$.

Question 10

If $\{X(t)\}$ and $\{Y(t)\}$ are independent WSS processes with zero means, find the ACF of $Z(t)$ when

(i) $Z(t) = a + bX(t) + cY(t)$, (ii) $Z(t) = aX(t)Y(t)$.

4.2 Hints

Conventions follow the Unit III notes: for a WSS process, $R(\tau) = E[X(t)X(t-\tau)]$, $S(\omega) = \int_{-\infty}^{\infty} R(\tau)e^{-i\omega\tau}d\tau$, and $R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)e^{i\omega\tau}d\omega$.

Hint 1

Use the working test for stationarity: $E[X(t)]$ and $\mathrm{Var}[X(t)]$ must both be constant. Here $f(y) = 1$, $0 < y < 1$, so $E(Y) = \int_0^1 y\,dy = \frac{1}{2}$ and $E(Y^2) = \int_0^1 y^2\,dy = \frac{1}{3}$, giving $\mathrm{Var}(Y) = \frac{1}{12}$. Since $\cos\omega t$ is not random,

$$E[X(t)] = \cos\omega t\, E(Y) = \tfrac{1}{2}\cos\omega t, \qquad \mathrm{Var}[X(t)] = \cos^2\omega t\, \mathrm{Var}(Y) = \tfrac{1}{12}\cos^2\omega t.$$

Both depend on $t \Rightarrow \{X(t)\}$ is **not** SSS (it is evolutionary).

See notes: SSS definition; Exercise on $A\cos(\omega_0 t + \theta)$, $\theta \sim U(0, \pi)$.

Hint 2

Let the common variance be σ^2. Then

$$E(Y) = E(Z) = 0, \quad E(Y^2) = E(Z^2) = \sigma^2, \quad E(YZ) = E(Y)E(Z) = 0 \text{ (indep.)}.$$

Mean: $E[X(t)] = E(Y)\cos\omega t + E(Z)\sin\omega t = 0$.
ACF: expand $R(t_1, t_2) = E\big[(Y\cos\omega t_1 + Z\sin\omega t_1)(Y\cos\omega t_2 + Z\sin\omega t_2)\big]$; the cross terms vanish and

$$\begin{aligned} R(t_1, t_2) &= \sigma^2(\cos\omega t_1\cos\omega t_2 + \sin\omega t_1\sin\omega t_2) \\ &= \sigma^2\cos\omega(t_1 - t_2). \end{aligned}$$

Constant mean and R a function of $t_1 - t_2$ only $\Rightarrow$ WSS.

See notes: Problem – $X(t) = Y\cos\omega t + Z\sin\omega t$ is a WSS process; Exercise with variance σ^2.

Hint 3

Separate real and imaginary parts of the characteristic function:

$$\begin{aligned} \phi(1) = 0 &\;\Rightarrow\; E(\cos Y) = 0,\; E(\sin Y) = 0; \\ \phi(2) = 0 &\;\Rightarrow\; E(\cos 2Y) = 0,\; E(\sin 2Y) = 0. \end{aligned}$$

Mean: $E[X(t)] = \cos\lambda t\, E(\cos Y) - \sin\lambda t\, E(\sin Y) = 0$.
ACF: with $\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$,

$$R(t_1, t_2) = \tfrac{1}{2}\cos\lambda(t_1 - t_2) + \tfrac{1}{2}E\big[\cos\big(\lambda(t_1 + t_2) + 2Y\big)\big].$$

Expand the last expectation as $\cos\lambda(t_1 + t_2)\, E(\cos 2Y) - \sin\lambda(t_1 + t_2)\, E(\sin 2Y) = 0$.
Hence $R(\tau) = \frac{1}{2}\cos\lambda\tau \Rightarrow$ WSS.

Hint 4

For a fixed t, $X(t)$ is a discrete random variable taking two values, each with probability $\frac{1}{2}$ (fair coin):

$$\begin{array}{c|cc} X(t) & \sin \pi t & 2t \\ \hline P & \frac{1}{2} & \frac{1}{2} \end{array} \qquad \Longrightarrow \qquad E[X(t)] = \tfrac{1}{2}\sin \pi t + \tfrac{1}{2}(2t) = \tfrac{1}{2}\sin \pi t + t.$$

(Note that the mean depends on t, so the process is not stationary.)

Hint 5

$f(\theta) = \frac{1}{2\pi}$, $-\pi < \theta < \pi$.

Mean: $E[X(t)] = \dfrac{10}{2\pi}\displaystyle\int_{-\pi}^{\pi} \cos(100t + \theta)\, d\theta = \frac{10}{2\pi}\big[\sin(100t + \theta)\big]_{-\pi}^{\pi} = 0.$

ACF: use the product-to-sum formula:

$$\begin{aligned} R(t_1, t_2) &= 100\, E\big[\cos(100t_1 + \theta)\cos(100t_2 + \theta)\big] \\ &= 50\cos 100(t_1 - t_2) + 50\, E\big[\cos\big(100(t_1 + t_2) + 2\theta\big)\big]. \end{aligned}$$

The last expectation integrates to 0 over a full period, so $R(\tau) = 50\cos 100\tau \Rightarrow$ WSS.

See notes: Problem 2 (radio station, $\cos(\omega_0 t + \theta)$); Problem – $A\cos(\omega_0 t + \theta)$ is a WSS process.

Hint 6

Use the two PSD properties proved in the notes.

- $R(\tau)$ is even, so $R(\tau)\sin\omega\tau$ is odd and drops out of $S(\omega) = \int R(\tau)(\cos\omega\tau - i\sin\omega\tau)\, d\tau$:
$$S(\omega) = 2\int_0^{\infty} R(\tau)\cos\omega\tau\, d\tau.$$
- $S(\omega)$ is even (Property 1), so in the inverse transform $R(\tau) = \frac{1}{2\pi}\int S(\omega)(\cos\omega\tau + i\sin\omega\tau)\, d\omega$ the sine part vanishes:
$$R(\tau) = \frac{1}{\pi}\int_0^{\infty} S(\omega)\cos\omega\tau\, d\omega.$$

These two relations form a Fourier cosine transform pair.

See notes: Properties of PSD – Properties 1 and 2; ACF Property 1.

Hint 7

$S(\omega) = \int_{-\infty}^{\infty} e^{-\alpha\tau^2} e^{-i\omega\tau}\, d\tau = \int_{-\infty}^{\infty} e^{-(\alpha\tau^2 + i\omega\tau)}\, d\tau$. Complete the square in the exponent:

$$\alpha\tau^2 + i\omega\tau = \alpha\Big(\tau + \frac{i\omega}{2\alpha}\Big)^2 + \frac{\omega^2}{4\alpha}.$$

Put $u = \tau + \frac{i\omega}{2\alpha}$ and use $\int_{-\infty}^{\infty} e^{-\alpha u^2} du = \sqrt{\pi/\alpha}$:

$$S(\omega) = \sqrt{\frac{\pi}{\alpha}}\, e^{-\omega^2/(4\alpha)}.$$

Second part: write $\cos\omega_0\tau = \frac{1}{2}\big(e^{i\omega_0\tau} + e^{-i\omega_0\tau}\big)$. Each term just shifts ω to $\omega \mp \omega_0$ in the result above, so

$$S(\omega) = \frac{1}{2}\sqrt{\frac{\pi}{\alpha}}\Big[e^{-(\omega-\omega_0)^2/(4\alpha)} + e^{-(\omega+\omega_0)^2/(4\alpha)}\Big].$$

See notes: PSD definition; PSD Problem 1.

Hint 8

$R(\tau) = \dfrac{1}{2\pi}\displaystyle\int_{-\infty}^{\infty} \frac{e^{i\omega\tau}}{(1+\omega^2)^2}\, d\omega = \frac{1}{\pi}\int_0^{\infty} \frac{\cos\omega\tau}{(1+\omega^2)^2}\, d\omega$ (integrand is even).
Evaluate by contour integration (double pole at $\omega = i$ for $\tau > 0$), or verify the answer directly by computing $2\int_0^{\infty} R(\tau)\cos\omega\tau\, d\tau$ with

$$\int_0^{\infty} e^{-\tau}\cos\omega\tau\, d\tau = \frac{1}{1+\omega^2}, \qquad \int_0^{\infty} \tau e^{-\tau}\cos\omega\tau\, d\tau = \frac{1-\omega^2}{(1+\omega^2)^2}.$$

You should get $R(\tau) = \frac{1}{4}(1 + |\tau|)\, e^{-|\tau|}$.
Average power: $R(0) = E[X^2(t)] = \dfrac{1}{2\pi}\displaystyle\int_{-\infty}^{\infty} \frac{d\omega}{(1+\omega^2)^2}$; put $\omega = \tan\theta$ to get $\frac{1}{2\pi}\cdot\frac{\pi}{2} = \frac{1}{4}$.
See notes: PSD – Inverse relation; PSD Problem 2.

Hint 9

Since R and ϕ are independent, the expectation splits:

$$R(t_1, t_2) = E\big[R^2\big]\; E\big[\cos(\omega t_1 + \phi)\cos(\omega t_2 + \phi)\big].$$

With $f(\phi) = \frac{1}{2\pi}$ on $(-\pi, \pi)$, the product-to-sum formula gives $E[\cos(\omega t_1 + \phi)\cos(\omega t_2 + \phi)] = \frac{1}{2}\cos\omega(t_1 - t_2)$ (the $\cos(\omega(t_1 + t_2) + 2\phi)$ term averages to 0). Hence

$$R(t_1, t_2) = \tfrac{1}{2}E(R^2)\cos\omega(t_1 - t_2).$$

See notes: Problem – $A\cos(\omega_0 t + \theta)$ is a WSS process.

Hint 10

Independence and zero means give $E[X(t)Y(t-\tau)] = E[X(t)]E[Y(t-\tau)] = 0$ (CCF Property 4: $R_{XY}(\tau) = \mu_X \mu_Y = 0$).
(i) $R_{ZZ}(\tau) = E\big[(a+bX(t)+cY(t))(a+bX(t-\tau)+cY(t-\tau))\big]$. Expand: terms with a single X or Y have zero mean, and the XY cross terms vanish, leaving

$$R_{ZZ}(\tau) = a^2 + b^2 R_{XX}(\tau) + c^2 R_{YY}(\tau).$$

(ii) $R_{ZZ}(\tau) = a^2 E\big[X(t)X(t-\tau)\,Y(t)Y(t-\tau)\big]$; by independence of the two processes,

$$R_{ZZ}(\tau) = a^2 R_{XX}(\tau)\, R_{YY}(\tau).$$

See notes: CCF – Property 4; Problem – CCF of $W(t)$ and $Z(t)$.